

REGULAR ARTICLE

How Does Complex Knowledge Promote Regional Economic Growth Within the “Belt and Road?”

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ABSTRACT

As economic globalization shifts toward regionalization, complex knowledge becomes increasingly crucial. Burgeoning literature begins to explore how complex knowledge facilitates economies. Given the representativeness of the “Belt and Road” Initiative in world economic regionalization, we extend this discussion within the countries involved, and make several distinctions. First, by combining the Fitness and Complexity (FC) Algorithm and the Matrix-Estimation exercise in the bipartite network, we develop a more comprehensive and effective method to measure knowledge complexity based on the similarity and specialization of patents’ output structure. Second, by employing the Instrument Variable Fixed Effect Quantile Regression (IV-FEQR), we investigate the economic effects of complex knowledge production while considering heterogeneity and endogeneity issues. Our findings indicate that increasing knowledge complexity significantly promotes a country’s economic growth, a result that remains robust across various checks. The benchmark result can be explained theoretically well through the macro- and micro-level driving forces, which are also empirically identified. Heterogeneity results further reveal that the promotion effect varies across regions and “expirations.” In addition, the results of moderating effect regressions also emphasize the positive role of the “Belt and Road” Initiative as a moderator; we also find that its interaction with external factors exhibits both a leveraging effect and a mitigating effect. Our analysis provides practical insights for measuring knowledge complexity and offers important policy implications for boosting economic growth amid current geopolitical tensions.

JEL Classification: D85, O47

1 | Introduction

As geopolitical tensions continue and economic globalization faces headwinds, international trade policy uncertainty substantially increases (Liu et al. 2023) and the traditional global collaboration mode based on comparative advantage and factor endowments has been impacted, followed by the emergence of regional trade organizations (Dutta 1992; Zhang et al. 2022), one of the most representative is the “Belt and Road” Initiative (Erbach 2021; Huang 2016). While regional trade organizations enhance economic ties among member countries and promote investment and resource sharing, they also significantly increase

the complexity of trade rules, leading to higher trade barriers in globalization (e.g., Spaghetti Bowl Effect (Sorgho 2016)). In this context, traditional Ricardian economic growth mode, which tends to treat technological progress as an exogenous factor, is inevitably challenged, prompting countries to focus more on innovation-driven development strategies powered by complex knowledge,¹ especially for developing countries with relatively lower technological level (Atkin and Khandelwal 2020) (e.g., China, India, Vietnam, etc.).

Logistically, the difficulty of learning or researching certain complex knowledge can induce high costs for individuals to enter the

corresponding field. In other words, the dissemination of such complex knowledge among economic entities will face substantial barriers (Balland and Rigby 2017; Sorenson et al. 2006). Therefore, if the complexity of knowledge is higher, such knowledge will be more exclusive and thus contain larger economic value (Mewes and Broekel 2022). Given the necessity of complex knowledge for economic growth in the developing world, we focus on the “Belt and Road” Initiative, which is mainly composed of developing countries (Erbach 2021; Huang 2016), and investigate the impact of complex knowledge on regional economic growth in the current process of economic de-globalization. We ask the following research questions: First, does complex knowledge play a positive role in facilitating economic growth within the “Belt and Road?” Second, if such a promotion effect exists, what is the underlying driving force? Third, does this promotion effect exhibit significant heterogeneity? What are the potential policy implications?

To address the above-mentioned questions, we start the empirical work by combining the Fitness and Complexity (FC) Algorithm proposed by Tacchella et al. (2013) and the matrix-estimation exercise introduced by Sciarra et al. (2018, 2020) in a bipartite network, and developing a more comprehensive and effective method to measure the knowledge complexity based on the similarity and specialization of patents’ outputting structure.² Then, leveraging the substantial variation in knowledge complexity across different countries and years within the “Belt and Road,” we develop the econometric model to investigate the causal relationship between knowledge complexity and economic growth. Our empirical work relies primarily on the item of Science, Technology, and Patents in the OECD statistics dataset, which can help develop the bipartite network and construct the knowledge complexity indicator. We also utilize various country-level indicators such as the economy, geography, and culture.

On the basis of the “Belt and Road” sample, which comprises data from 41 countries from 1990 to 2021, the results of baseline regressions with a series of robustness checks demonstrate a significantly positive relationship between knowledge complexity and regional economic growth within the “Belt and Road.” A result that is in line with the idea that technological progress is a decisive factor for economic growth (Romer 1986), especially for developing countries in the context of de-globalization. Then, this article identifies the macro-economic and micro-economic channels as two underlying mechanisms through which knowledge complexity boosts regional economic growth; and at the same time, this article also conducts heterogeneous analyses regarding the geographic aspect and “expirations” of patents, and sensitivity analyses of heterogeneous promoting effect across different technology domains and economic sectors. Final, we extend our research sample to include 100 countries worldwide and explore the moderating effect of the “Belt and Road” Initiative and its interactions with external factors. The results highlight the reinforcing role of the “Belt and Road” Initiative. Moreover, the advance of the BRI further strengthens the positive effects arising from political alliances and geopolitical stability, while also helping to offset the adverse impacts of de-globalization and trade protectionism.

The remainder of the article is organized as follows. Section 2 briefly reviews the related literature, details the research gap, and proposes the research hypotheses. Section 3 introduces the

measurement method of the key independent variable of interest—knowledge complexity, and discusses the evolution patterns of knowledge complexity in the “Belt and Road,” providing stylized facts for the following empirical work. Section 4 outlines the econometric model and variable specifications. Section 5 presents the regression analyses, reporting the baseline results, along with a series of robustness checks, mechanism tests, and heterogeneity analyses. Section 6 extends the sample to include non-BRI countries and further examines the moderating role of the BRI, as well as its interactions with external factors. Section 7 concludes the article.

2 | Literature Review and Hypotheses

2.1 | Related Literature

Our study is closely related to the burgeoning literature on the economic effect of complex knowledge, which covers various fields. For individual employment, Ganco (2013) examines the effect of the complexity of employees’ prior patenting activities on their decisions to join a rival firm or found a start-up and finds that complexity can inhibit creative employees from moving to rival firms but can promote employee entrepreneurship. Eriksson and Forslund (2014) find that employees characterized by the synthetic knowledge base can better match the knowledge produced by universities and realize higher employment growth. For industrial upgrading, Feng and Liu (2018) argue that knowledge development, absorption, integration, and creation are crucial for the internal upgrading of the industry and enhancing economic complexity. As for a specific industry, Bai et al. (2020) study the industrial upgrading effect of “quid pro quo” (technology for market access) in the Chinese automobile industry and find that foreign technology spillovers play a key role in domestic industrial upgrading, thereby promising more effective industrial linkages (Surana et al. 2020). For innovation, scholars generally believe that complex knowledge can further catalyze greater innovation through the channel of “learning by doing” (Garavito-Bermúdez 2018; Pham et al. 2021), creating a more pressurized innovation environment (Ruoslahti 2020) and accelerating new knowledge accumulation (Audretsch and Belitski 2021).

From a global economic perspective, complex knowledge-induced employment, industrial upgrading, and innovation in the domestic market will further enhance one region’s competitiveness in the international market (Feng and Liu 2018; Tacchella et al. 2013), and in turn, realize long-term economic growth (Arthur 1999; Audretsch and Belitski 2021; Balland et al. 2019; Helbing and Kirman 2013). Although existing literature has comprehensively discussed the economic effects of complex knowledge, a majority of it only considers the “yield” of complex knowledge without considering the “density” of complex knowledge based on the heterogeneity of knowledge and the correlation between research objects. In this article, we extend this brunch of literature by combining the FC Algorithm proposed by Tacchella et al. (2013) and the matrix-estimation exercise introduced by Sciarra et al. (2018, 2020) in a bipartite network, thereby developing a more comprehensive and effective method to measure the knowledge complexity based on the similarity and specialization of patents’ outputting structure,

holding significant practical value in both the application of complex network theory and research of innovation and knowledge economics.

Our article is also related to the literature on regional trade organizations, especially for the “Belt and Road.” With the current trend in the global economy development moving toward regionalization (Dutta 1992), an increasing number of studies are beginning to focus on the “Belt and Road” Initiative, covering various fields such as scientific collaboration, knowledge spillover, tourism, energy development, and environment (Hashemizadeh et al. 2021, 2024; Pan and Hashemizadeh 2023). In this article, we supplement this stream of literature and explore the impact of complex knowledge on economic growth within the “Belt and Road.” In addition, for causal inference, we employ the Instrument Variable-Fixed Effect Quantile Regression (IV-FEQR) method to better address the endogeneity issue, enhancing the reliability of our empirical findings.

2.2 | Research Hypotheses

Technological progress has long been a crucial engine of economic growth (Romer 1986). In the current context of de-globalization, the optimal mode of economic cooperation among countries may be limited, and innovation-driven development strategies based on complex knowledge are receiving increasing attention. Since the substantial cost in the dissemination of complex knowledge (Balland and Rigby 2017; Sorenson et al. 2006), the economic value of complex knowledge is very evident (Mewes and Broekel 2022). Therefore, the role of complex knowledge in driving economic growth has become a widely accepted consensus in related literature (Arthur 1999; Audretsch and Belitski 2021; Balland et al. 2019; Helbing and Kirman 2013), especially for developing countries. Thus, the following hypothesis will be tested:

Hypothesis 1. *The increase in knowledge complexity can promote regional economic growth within the “Belt and Road”.*

Through the previous review of existing literature, we can conclude, from a macroeconomic perspective, that: (1) Increased knowledge complexity enhances international competitiveness. A higher level of knowledge complexity indicates that the economy is better equipped to produce complex knowledge outputs, which is evident in the production of high-tech products (Tallman et al. 2004). Mastery of complex knowledge allows the economy to engage in critical points of the value chain, providing more substantial competitive advantages in the international market and enabling the realization of more excellent benefits (Feng and Liu 2018; Lagerström et al. 2023). (2) The production of complex knowledge may lead to more effective industrial linkages and provide aid in optimizing and upgrading industrial structure. In fact, promoting knowledge complexity supports industrial structural upgrading by aggregating innovative elements and enhancing the capacity to absorb and transform technological achievements indirectly (Feng and Liu 2018; Innocenti et al. 2020). Moreover, on the other hand, as critical participants in intermediate inputs and manufacturing outputs, sectors with higher knowledge complexity can also play a more

pivotal role in promoting the transformation and upgrading of domestic industrial structures through continuous technology updates and iterations (Bai et al. 2020). Furthermore, they can enhance operational efficiency and talent reserves, thereby driving economic growth (Surana et al. 2020). Thus, the following hypothesis will be tested:

Hypothesis 2a. *By strengthening international competitive advantages, the increase in knowledge complexity can drive regional economic growth.*

Hypothesis 2b. *By enhancing industrial linkages and optimizing industrial structure, the increase in knowledge complexity can also lead to regional economic growth.*

From a microeconomic perspective, particularly innovational practice, we can conclude that: (1) Higher knowledge complexity fosters a more pronounced “learning by doing” effect, allowing firms to further accumulate new knowledge and constantly achieve technological innovation (Audretsch and Belitski 2021; Bai et al. 2020; Garavito-Bermúdez 2018; Pham et al. 2021). (2) The increase in knowledge complexity creates a more pressurized innovation environment, continuously stimulating the motivation for technological innovation (Ruoslahti 2020), and ultimately incentivizing innovation and technological accumulation. The scale and structure of generated complex knowledge not only reflect the local production capacity and development potential but also drive technological progress through collaborative innovation processes, thereby promoting economic growth (Fischer and Frhlich 2010; Ruoslahti 2020). Thus, the following hypothesis will be tested:

Hypothesis 3a. *The increment in knowledge complexity can promote regional economic growth through innovational practices such as “learning by doing”.*

Hypothesis 3b. *The increase in knowledge complexity can also drive regional economic growth through technological accumulation.*

As for the “Belt and Road” Initiative, economic ties among member countries are substantially enhanced; the majority of the members are developing countries with lower technological levels (Erbach 2021; Huang 2016), these countries often have greater untapped resources and markets, relatively weak infrastructure, and the introduction of complex technology can lead to significant improvements and efficiency gains. Additionally, the markets in developing countries are not yet saturated, allowing technological advancements to more effectively meet demand and promote consumption; the promotional effect of complex knowledge can play a pronounced role in the “Belt and Road” countries. Thus, the following hypothesis will be tested:

Hypothesis 4. *The “Belt and Road” Initiative has a significantly positive moderating effect on the relationship between complex knowledge production and economic growth.*

In the process of international knowledge transfer and technological cooperation, political proximity is often associated with a country’s willingness and ability to engage in cooperation, receive foreign aid, and access development resources (Aiyar

et al. 2024; Bracco et al. 2015). Nonetheless, the role of political proximity may be limited in the absence of formalized institutional mechanisms, without which the benefits of cooperation are often diminished and the risk of breakdowns increases. Given its strategic orientation, the BRI provides a more structured platform for resource allocation and policy coordination, thereby enhancing the efficiency with which knowledge complexity is translated into economic growth.

Arising from international conflicts or border disputes, Geopolitical risk, which is widely recognized as a key external shock to economic performance (Busse and Hefeker 2007; Carney 2016), can interrupt infrastructure development and impede the transmission of technological capabilities (Lee et al. 2023). For knowledge-intensive industries in particular, the predictability of the political environment is a necessary condition for sustained innovation, which, in turn, depends on stable investment in knowledge and human capital (Tebaldi 2011). As a broad and influential multilateral initiative, the BRI may help participating countries establish more stable institutional expectations for knowledge diffusion and innovation, thereby allowing complex knowledge to more fully realize its economic growth potential.

Amid the current pushback against globalization, trade barriers are increasingly seen as major impediments to the cross-border flow of knowledge and technology (Autor et al. 2020). While countries face higher institutional barriers in external economic engagement, their capacities to absorb complex knowledge are greatly diminished. By investing in infrastructure and facilitating cross-border factor mobility, the BRI aims to promote regional economic integration, which in turn reduces the risks of economic and technological “decoupling” among countries, thereby partially offsetting the negative effects of de-globalization on the growth-enhancing role of complex knowledge.

Thus, the following hypothesis will be tested:

Hypothesis 5a. *Closer political alignment strengthens the positive moderating effect of knowledge complexity, and the “Belt and Road” Initiative further amplifies such a moderating role.*

Hypothesis 5b. *A stable geopolitical environment enhances the growth effects of knowledge complexity, and the BRI further reinforces the role of geopolitical stability.*

Hypothesis 5c. *De-globalization and trade protectionism weaken the growth-enhancing effects of knowledge complexity, while the BRI helps mitigate the negative impacts associated with de-globalization.*

3 | Measuring Knowledge Complexity

Before formally investigating the impact of knowledge complexity on economic growth, we first need to conduct quantification on complex knowledge, in order to measure the complexity of knowledge contained within an entity, which is often derived from patent data. In fact, patent output data have been a staple

of a considerable amount of macro-level literature on knowledge production (Jaffe and Trajtenberg 2002; Kogler et al. 2015; Ó hUallcháin 1999).

3.1 | Bipartite Network Framework

In previous research, many attempts have been made to measure the complexity of knowledge (Fleming and Sorenson 2001; Goldreich and Petrank 1999; Pintea and Thompson 2007). However, as mentioned earlier, these approaches mostly rely on simple calculations of knowledge inputs and outputs, such as the number of patents, citation data, and the number of knowledge domains combined in patents (Ejermo 2009; Schoenmakers and Duysters 2010; Verspagen 2007), without evaluating the intrinsic properties of the knowledge itself. Moreover, the subjects of knowledge complexity studies are often isolated from each other, ignoring the relational properties between subjects. It means that such measurements lack consideration of both the “heterogeneity” and “interconnectedness” of knowledge, making them less convincing (Balland and Rigby 2017).

In fact, measuring knowledge complexity first requires considering the heterogeneity of knowledge—different domains of knowledge contribute differently to the complexity of an entity. It involves evaluating the differences in the intrinsic properties of knowledge from other domains. Additionally, interconnectedness must be included in the measurement framework—changes in the complexity attributes of one or even multiple knowledge domains can affect the complexity of knowledge acquired or mastered by an economic entity, which in turn influences those of other knowledge domains; and the true for economic entities. Therefore, following the idea of Balland and Rigby (2017), Zhang and Zhang (2022), and Zhang (2024), this article aims to measure knowledge complexity by constructing a bipartite network framework. Here, we briefly introduce the method for constructing the knowledge complexity indicators, with more specific construction details available in the [Supporting Information](#).

First, constructing a bipartite network requires mapping network nodes and edges to patent data. Given the analytical background of this article, the two types of nodes in the bipartite network correspond to country nodes c and technology domain nodes t ; the edges correspond to the patent output M_{ct} of different countries in different technology domains. In the bipartite network, nodes of two different types have their own “centrality”, and the complexity of the knowledge possessed by countries and technology domains— X_c and Y_t , corresponds to some kind of “centrality” which can be calculated through certain algorithms. It is important to note that X_c and Y_t here refer to the “properties” of nodes, so this article terms them “knowledge compliances” of countries and technology domains, distinguishing them from the “knowledge complexity” mentioned later; “knowledge complexity” is actually the quantitative result of X_c and Y_t through an algorithm. As previously mentioned, due to the “heterogeneity” and “interconnectedness” of knowledge, X_c depends on $Y_1, Y_2, \dots, Y_t$, and M_{ct} ; similarly, Y_t depends on $X_1, X_2, \dots, X_c$, and M_{ct} . Thus, the problem of measuring knowledge complexity can just be equal to solving the following coupled equations:

$$\begin{cases} X_c = f(Y_1, Y_2, \dots, Y_t, M_{ct}) \\ Y_t = g(X_1, X_2, \dots, X_c, M_{ct}) \end{cases} \quad (1)$$

Then, calculating knowledge complexity requires constructing an algorithm based on the relationships between two types of nodes. In fact, it is not difficult to be aware of the relationship: for a country, the more technological domains and the higher the complexity of the technological domains it participates in, and the deeper its participation in highly complex technological domains, the higher the complexity of knowledge it will possess. Conversely, for a technological domain, if it is participated in by more countries, the knowledge complexity of it should be lower. Therefore, this article adopts the FC algorithm proposed by Tacchella et al. (2013) to describe the relationship between nodes of countries and technological domains, with the specific expression as follows:

$$\begin{cases} \tilde{F}_c^{(0)} = \sum_t M_{ct} Q_t^{(n-1)}, & \forall c \\ \tilde{Q}_t^{(0)} = \frac{1}{\sum_c M_{ct} \frac{1}{\tilde{F}_c^{(n-1)}}}, & \forall t \end{cases} \Leftrightarrow \begin{cases} F_c^{(n)} = \frac{\tilde{F}_c^{(n)}}{\sum_c \tilde{F}_c^{(n)} / C} \\ Q_t^{(n)} = \frac{\tilde{Q}_t^{(n)}}{\sum_t \tilde{Q}_t^{(n)} / T} \end{cases} \quad (2)$$

where F_c denotes fitness for countries, Q_t denotes complexity for technologies, and the initial conditions are: $\begin{cases} \tilde{F}_c^{(0)} = 1, & \forall c \\ \tilde{Q}_t^{(0)} = 1, & \forall t \end{cases}$.

Through a series of mathematical transformations, and let $\begin{cases} X_c = F_c / s_c \\ Y_t = Q_t / s_t \end{cases}$, we can obtain two proximity matrixes— $\mathbf{N}$ and $\mathbf{G}$:

$$\begin{cases} N_{cc'} = \frac{\sum_{ct} M_{ct} M_{ct'}}{(s'_t)^2 s'_c s_c} \\ G_{tt'} = \frac{\sum_{ct} M_{ct} M_{ct'}}{(s_t)^2 s'_t s'_t} \end{cases} \quad (3)$$

where $\mathbf{N}$ can be explained as the proximity matrix of the country, describing the similarity of the structure of countries' participation in technology domains; $\mathbf{G}$ can be correspondingly explained as the proximity matrix of the technology, describing the similarity of the structure of technology domains' being participated in by countries. Thereinto, $s_c = \sum_t M_{ct}$, $s_t = \sum_c M_{ct}$, and $s'_t = \sum_c M_{ct} / s_c$.

Finally, using the Matrix-estimation exercise proposed by Sciarra et al. (2018), the knowledge complexity of countries and technological domains, for example, GPYC (Generalized knowledge-complexityY of Country) and GPYT (Generalized knowledge-complexityY of Technology domain), can be constructed as follows:

$$\begin{cases} GPYC_c = \left(\sum_{i=1}^2 \lambda_i^N (v_{c,i}^N)^2 \right)^2 + 2 \sum_{i=1}^2 (\lambda_i^N)^2 (v_{c,i}^N)^2 \\ GPYT_t = \left(\sum_{i=1}^2 \lambda_i^G (v_{t,i}^G)^2 \right)^2 + 2 \sum_{i=1}^2 (\lambda_i^G)^2 (v_{t,i}^G)^2 \end{cases} \quad (4)$$

where $\lambda_1^N / \lambda_2^N$ and $\lambda_1^G / \lambda_2^G$ are the largest 2 eigenvalues of the proximity matrixes $\mathbf{N}$ and $\mathbf{G}$, $v_{c,1}^N / v_{c,2}^N$ and $v_{t,1}^G / v_{t,2}^G$ are the eigenvectors corresponding to $\lambda_1^N / \lambda_2^N$ and $\lambda_1^G / \lambda_2^G$, respectively. Worth mentioning, $\mathbf{N}$ and $\mathbf{G}$ here are the modified matrix, by setting all diagonal elements to an arbitrary constant value (such as 0), namely $\begin{cases} N_{cc'} = \frac{\sum_{ct} M_{ct} M_{ct'}}{(s'_t)^2 s'_c s_c}, & \forall c' \neq c \\ N_{cc'} = 0, & c' = c \end{cases}$ and $\begin{cases} G_{tt'} = \frac{\sum_{ct} M_{ct} M_{ct'}}{(s_t)^2 s'_t s'_t}, & \forall t' \neq t \\ G_{tt'} = 0, & t' = t \end{cases}$.

The higher the $GPYC/GPYT$ is, the higher the knowledge complexity of a country or technology will be.

3.2 | Data Processing

The concept of “technology vertex” and “country vertex” mentioned in this article is based on the country-technology (domain) bipartite network, which is constructed by the data on the patent outputting of each technology domain from each country. The data is collected from the item of Science, Technology and Patents in the OECD statistics dataset, which encompasses the number of patents output each year of 100 countries (or regions) from 1985 to 2021, including detailed states of 35 sub-domains classified by WIPO.

The country-technology bipartite network constructed in this article is based on the RCA (Revealed Comparative Advantage) proposed by Balassa (Balassa 1965), which means that the element M_{ct} of the Adjacency Matrix $\mathbf{M}$ corresponding to the bipartite network is actually the RCA_{ct} . The RCA_{ct} can be calculated from the data mentioned above through the following formula:

$$M_{ct} = RCA_{ct} = \frac{P_{ct} / \sum_t P_{ct}}{\sum_c P_{ct} / \sum_t \sum_c P_{ct}} \quad (5)$$

However, it is worth noticing that, given the property of knowledge included by the patent, since a patent completes to be “output”, such a patent will be valid for a long term; or in other words, it will not expire within a certain spell, providing a contribution to the knowledge complexity of the technology domain corresponding. Therefore, while calculating the RCA of a certain year, this article counts the number of patents within a certain period before this year, not just the number of patents for this year.

On account of that, this article will conduct two parts of analyses based on the “expiration” of patents. One is based on the hypothesis of patents being valid in the long term, which means that the RCA will be calculated by using the number of patents from 1985 to a certain year, namely $P_{ct,y} = \sum_{i=1985}^y N_{ct,i}$. The other is based on the hypothesis of patents expiring after certain years, meaning that the RCA will be calculated by: $P_{ct,y} = \sum_{i=y-exp}^y N_{ct,i}$, where exp. denotes “expiration.”

3.3 | Results and Discussion

In terms of the calculation results of knowledge complexity, this article does not aim to provide lengthy and tedious

TABLE 1 | GPYC results of countries or regions within the “Belt and Road” in 2021.

GPYC ranking	Country or region	GPYC	Patents	Patents ranking
2	China	0.99778	63202.2	1
3	Singapore	0.93718	1713.8	17
4	Taiwan, PRC	0.91739	406.6	29
6	Malaysia	0.86690	136.2	42
10	Israel	0.78009	2076.6	14
12	Pakistan	0.70634	0.5	96
13	Hong Kong, PRC	0.69830	817.5	25
17	Belarus	0.67661	12.3	71
24	India	0.60700	1985.1	15
25	Estonia	0.60416	47.3	51
26	Lebanon	0.60363	5.8	81
27	Russia	0.57890	899.8	24
28	Saudi Arabia	0.57138	789.1	26
30	Thailand	0.56451	147.7	41
31	Romania	0.54873	27.5	65
35	Czechia	0.52963	277.9	33
36	Hungary	0.52904	102.7	45
38	Philippines	0.52648	28.9	63
39	Jordan	0.52569	23.0	66
40	Poland	0.52139	353.1	31

Note: GPYCs have been min–max normalized.

statements and analyses, but instead focuses on the discussion involving the “Belt and Road” countries. First, Table 1 shows results of some “Belt and Road” countries with their GPYCs; and Figure 1 shows changes of GPYC and GPYT rankings from 1990 to 2021. The full calculation results of GPYTs and GPYCs are provided in the Supporting Information Tables S1 and S2.

From the view of GPYCs, China is ranked 2nd and Taiwan (PRC), as one of the “Four Asian Tigers,” is ranked 4th; both are considered to be high-tech countries and regions. Meanwhile, countries or regions that rank lower are mostly developing countries in central or Southeast Asia with poorer development. However, contrary to common perception, developing countries also have high GPYCs, even though they were not previously considered to have high levels of technological advancement, for example, Malaysia and Pakistan (ranked 6th and 12th respectively); but countries that are considered to have the potential for higher technological development just do not have high GPYCs, for example, India (ranked 24th). In addition, it is not difficult to see from the table that there is no explicit correlation between GPYC and the volume of patent outputting.

Then, in order to furtherly analyze and explain the calculation results obtained above, this article maps the two largest eigenvectors $\mathbf{v}_1^N/\mathbf{v}_2^N$ corresponding to the proximity matrix $\mathbf{N}$ onto the x- and y-coordinate, respectively. Results of 100 countries (or regions) in 2021 are shown in Figure 2. In addition, this article also calculates all results of countries' scattering in $\{\mathbf{v}_1^N, \mathbf{v}_2^N\}$ plane from 1990 to 2021, which are shown in Figure 3; thereinto, trajectories of some countries are marked.

Actually, two indicators— $\mathbf{v}_1^N$ and $\mathbf{v}_2^N$ —both have clear topological and economic meanings; the former denotes the Eigenvector Centrality of the weighted undirected network

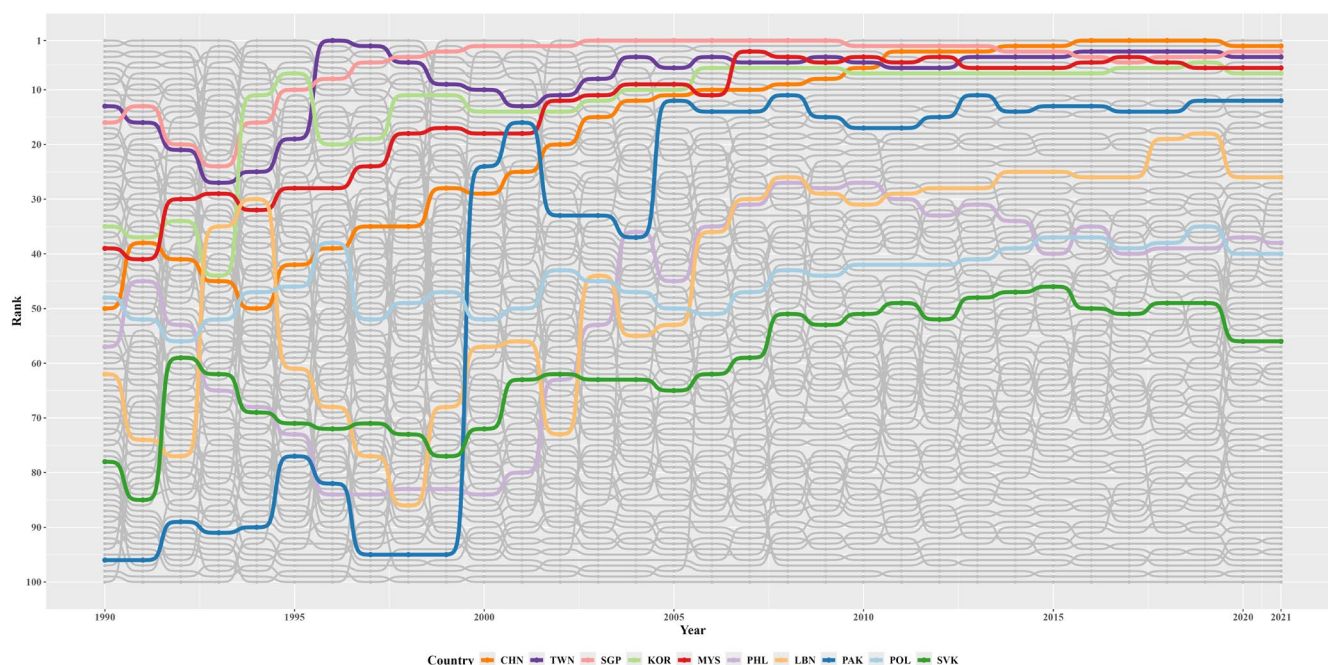


FIGURE 1 | Ranking changes of GPYC in 1990–2021. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/rdev.12056)]

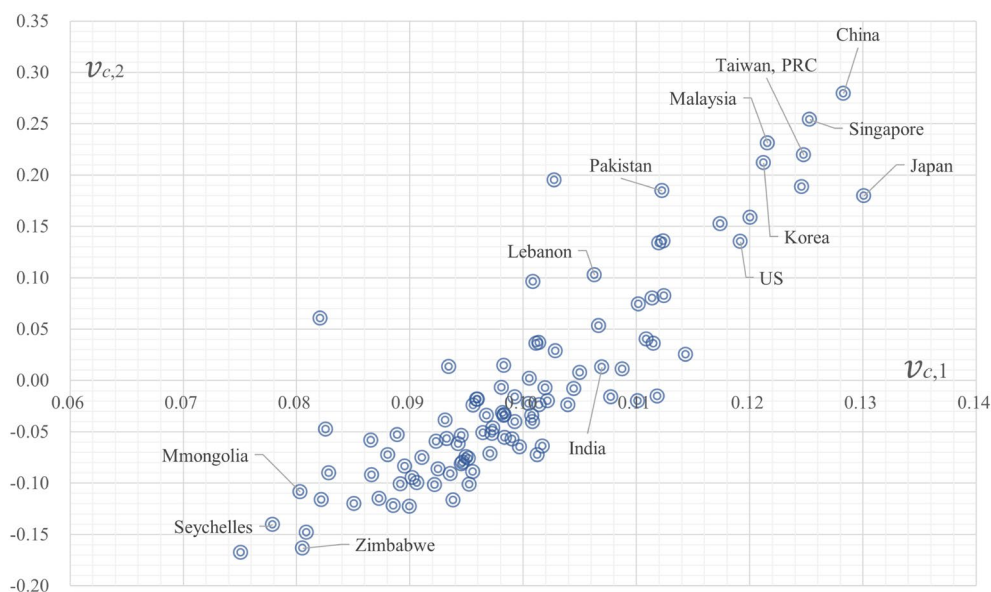


FIGURE 2 | Scatter plots on $\{v_1^N, v_2^N\}$ plane in 2021. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

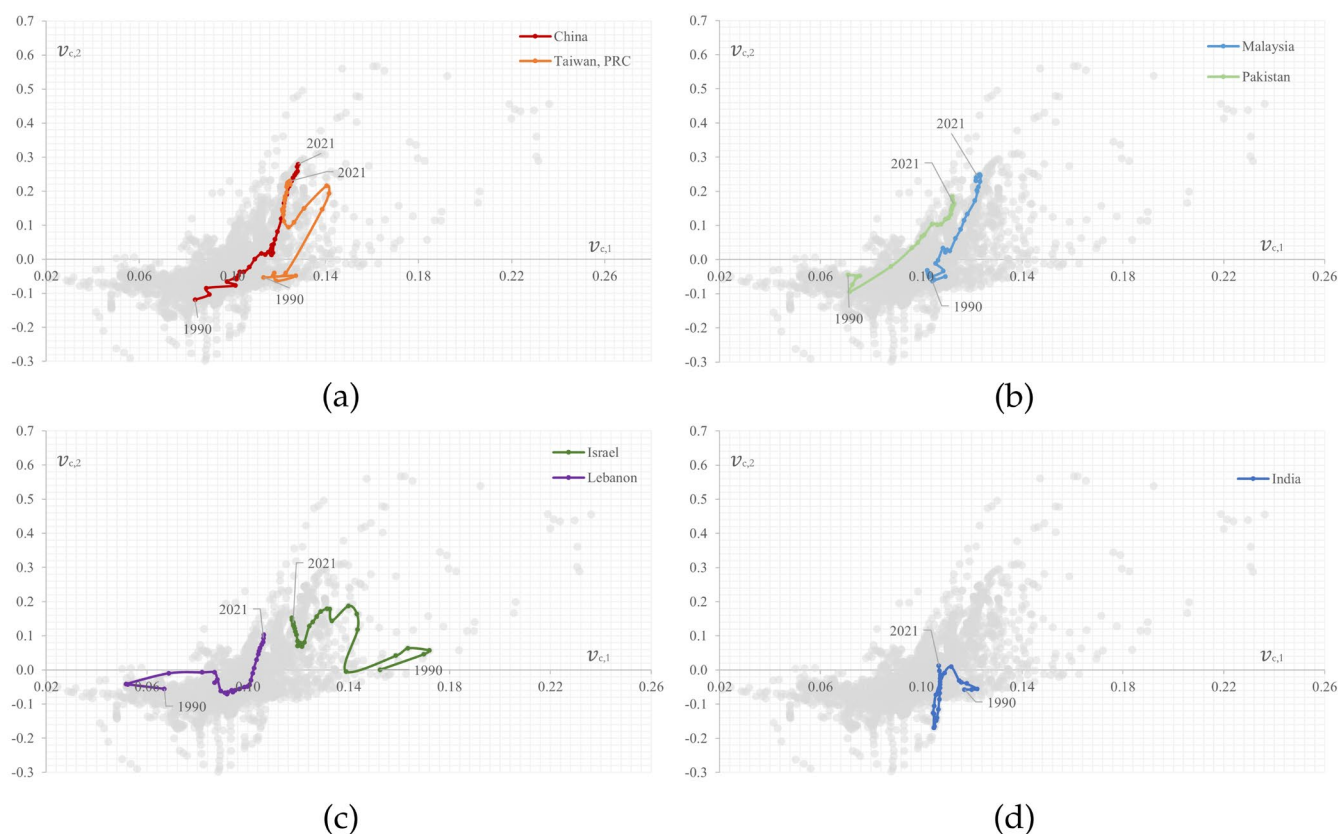


FIGURE 3 | Trajectories on $\{v_1^N, v_2^N\}$ plane from 1990 to 2021. (a) shows the Trajectories of China and Taiwan (PRC); (b) shows Malaysia and Pakistan; (c) shows Israel and Lebanon; (d) shows India. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

corresponding to and N (Hidalgo and Hausmann 2009; Tacchella et al. 2013), while the latter denotes the clustering information embedded in the topological structure of the corresponding network (Mealy et al. 2019; Sciarra et al. 2020). They respectively correspond to the “similarity” and “specialization” of countries in patent output (Zhang 2024), a higher value of $v_{c,1}^N$ means that the corresponding country tends to

have more similarity on patent contribution to countries with higher GPYC; and a higher value of v_2^N indicates that the corresponding country has a higher degree of specialization in a certain technology domain. In addition, if $v_{c,2}^N$ for a certain country is a positive value, this country tends to have higher specialization in technology domains with higher knowledge complexity, and vice versa.

From Figure 3, we can clearly understand the reasons why some countries unusually possess high GPYCs and their changes in knowledge complexity. Here, we select three countries for brief analysis. First, we take Israel—which has a relatively high knowledge complexity—as an example. The two domains with the highest proportion of patent outputting are medical technology and computer technology, as shown in Table 2. As mentioned above, computer technology has a high GPYT, while medical technology, although with a lower GPYT, is highly specialized in Israel. Therefore, the GPYC of Israel is not low. It is also worth mentioning that a large portion of the fields that are top 10 patent-producing in Israel are domains with high GPYT, for example, digital communication, telecommunications, optics, etc.; and compared with other “Belt and Road” countries, Israel's outputting in such high-techs remains at a higher level generally. It can be attributed to the horizontal technology policies (HTPs) around the 1990s, when Israel launched catalytic programs that spurred private technological entrepreneurship, playing a crucial role in the growth of key companies (Breznitz 2007; Teubal 1999).

Although Pakistan has indeed been a country with relatively low levels of technological development for a long time, its level of technological development has been upgraded rapidly in the past decade or so. Particularly in the fields of audio-visual technology, computer technology, and digital communication, patent output data provided by the OECD, as shown in Table 3, indicates that the country's patent output in such two fields began to grow dramatically after 2010, far exceeding other domains. Although it is limited in terms of its total patent output in all technology domains, such a “specialization” has allowed Pakistan to leverage its “advantage” in high knowledge complexity technologies. It can be attributed to the introduction of Pakistan's first comprehensive policy on science and technology, the “National Science, Technology, and Innovation Policy 2012,” which has significantly advanced Pakistan's capacity for technological innovation (Bhutto et al. 2012; Ul-Haq et al. 2014).

Finally, looking at the situation in India, although its development of science and technology is considered to be leading, its participation across various technological domains is relatively

TABLE 2 | Patents outputting of Israel.

Technology domain	1990	1995	2000	2005	2010	2015	2021	Proportion (2021)
Medical technology	2.50	52.62	507.20	1446.25	2820.94	4287.23	6462.27	17.72%
Computer technology	1.00	21.72	248.05	801.93	1794.69	2509.75	3707.51	10.17%
Pharmaceuticals	2.26	61.70	317.95	905.77	1667.05	2315.77	3393.43	9.30%
Measurement	0.00	29.57	230.37	569.21	919.77	1305.36	2055.96	5.64%
Biotechnology	1.61	48.77	239.06	573.61	949.42	1292.49	1765.46	4.84%
Digital communication	0.00	2.95	146.50	460.52	713.47	993.79	1385.26	3.80%
Telecommunications	0.50	12.04	178.22	493.60	836.72	1039.86	1219.36	3.34%
Other special machines	0.33	12.93	114.12	282.87	477.28	698.66	1154.32	3.16%
Optics	1.00	24.84	130.82	345.90	510.25	699.65	1077.63	2.95%
Electrical machinery, apparatus, energy	0.00	17.94	137.30	300.58	532.51	772.29	1059.35	2.90%

Source: OECD statistics and the authors' calculations.

TABLE 3 | Patents outputting of Pakistan.

Technology domain	1990	1995	2000	2005	2010	2015	2021	Proportion (2021)
Audio-visual technology	0.00	0.00	0.00	0.00	4.67	5.33	6.17	19.87%
Computer technology	0.00	0.00	0.00	0.00	0.17	1.25	3.88	12.49%
Digital communication	0.00	0.00	0.00	0.00	0.00	0.92	3.67	11.82%
IT methods for management	0.00	0.00	0.00	0.00	0.25	0.25	2.96	9.53%
Furniture, games	0.00	1.00	1.00	1.00	1.00	1.58	2.08	6.71%
Telecommunications	0.00	0.00	0.00	0.00	0.50	0.67	1.75	5.64%
Pharmaceuticals	0.00	0.00	0.00	0.00	1.00	1.00	1.61	5.19%
Basic materials chemistry	0.00	0.00	0.00	0.00	0.00	0.67	1.39	4.48%
Medical technology	0.00	0.00	0.00	0.00	0.55	0.88	1.38	4.44%
Basic communication processes	0.00	0.00	0.00	0.00	1.00	1.00	1.00	3.22%

Source: OECD statistics and the authors' calculations.

TABLE 4 | Patents outputting of Inida.

Technology domain	1990	1995	2000	2005	2010	2015	2021	Proportion (2021)
Pharmaceuticals	0.40	0.40	112.68	930.04	2104.76	3504.43	5052.74	20.17%
Organic fine chemistry	0.00	0.05	62.40	931.51	2096.41	3332.18	4355.92	17.39%
Computer technology	0.00	0.00	3.68	101.36	308.31	685.54	1489.69	5.95%
Medical technology	0.00	0.00	6.25	62.51	209.80	494.77	1257.01	5.02%
Biotechnology	0.60	0.85	18.47	199.86	476.70	805.49	1184.56	4.73%
Basic materials chemistry	0.00	0.00	17.11	111.74	275.88	541.17	984.46	3.93%
IT methods for management	0.00	0.00	3.78	22.74	146.90	382.32	923.59	3.69%
Transport	0.00	1.00	3.00	36.28	168.54	321.53	859.98	3.43%
Electrical machinery, apparatus, energy	0.00	0.00	5.33	39.70	138.87	307.03	639.69	2.55%
Chemical engineering	0.00	0.00	4.39	70.77	167.94	383.26	638.16	2.55%

Source: OECD statistics and the authors' calculations.

“dispersed,” similar to that of the United States. It is well known that India, benefiting from the industrial chain transformation of developed countries, is located in the leading place worldwide, referring to pharmaceuticals and organic chemistry (Akhtar 2013). However, these two technologies do not possess high knowledge complexity, and overall, India also exerts a relatively low concentration in high knowledge complexity fields. As shown in Table 4, it is not difficult to find out that, in terms of patent output in 2021, a significant portion of the top 15 technology domains is in low knowledge complexity. The dispersed and widespread participation across various technologies has led to its knowledge complexity not being high.

In fact, the knowledge complexity calculated in this article measures the “intensity” of complex knowledge output, which is based on “outperformance” in patent outputs referring to specific technologies. It means that countries with higher knowledge complexity have stronger R&D capabilities in certain fields, capable of producing more knowledge-intensive products, shaping “industrial advantages,” and thereby creating higher economic value. It also means that knowledge complexity depends on the degree of similarity in patent contribution structures with other countries and the degree of specialization in various technology domains, rather than being related to the total amount of knowledge output. Moreover, the above conclusion implies a policy recommendation: achieving the “division of labor” within technology domains is undoubtedly a good way to enhance the level of scientific and technological development. For different countries, it is advisable to choose which knowledge fields to specialize in based on their actual development conditions and factor endowments.

4 | Research Design

4.1 | Econometric Model and Variable Specifications

Formally, this article considers the use of the following regression to examine how economic growth varies with knowledge complexity:

$$econ_growth_{it} = \alpha + \beta tech_level_{it} + \sum_k^K \gamma_k x_{k,it} + \lambda_t + \mu_i + \varepsilon_{it} \quad (6)$$

In Equation (6), *econ_growth* denotes the explained variable, namely, economic growth; *tech_level* denotes the core explanatory variable, namely, technological level. $\sum_k^K \gamma_k x_{k,it}$ denotes control variables; λ_t and μ_i denotes the time and individual fixed effects respectively. In fact, the regression model expressed by Equation (6) is a generalized form; by employing different estimation approaches, different coefficients of the core explanatory variable *GPYC_rank* (namely $\hat{\beta}$) can be obtained. In this article, we consider conducting regressions by employing Instrument Variable Fixed Effect Quantile Regression (IV-FEQR) as the estimation method.

This article employs two indicators as the core explanatory variable (corresponding to *econ_growth*), namely, GPYC_rank and GPYC. As elaborated in previous sections, this article aims to quantify a country's scientific and technological capacity by leveraging patent data and employing a specific algorithm to calculate knowledge complexity. Section 3 details the construction of the GPYC indicator and presents cross-national estimates of knowledge complexity. To facilitate the interpretation of coefficients in subsequent regression analyses, the article recalculates the ranking percentile of the GPYC value of each country or region, denoted as GPYC_rank. This variable serves as the primary explanatory variable in both the baseline regressions and various robustness checks. Additionally, to ensure the robustness of the findings, the original GPYC values are also used as the core explanatory variable in robustness checks.

Nine different indicators are employed as the explained variable (corresponding to *tech_level*). First, two leading indicators, namely, Gross Domestic Product (GDP) per capita growth and Gross National Income (GNI) per capita growth, are used. As conventional macroeconomic indicators, the two indicators are widely recognized as the most direct and widely accepted measures of economic growth and macroeconomic output, offering

a reliable benchmark relative to more indirect proxies. Second, this article incorporates household consumption and CO₂ emissions based on consumption to reflect both household and aggregate consumption behaviors. The two indicators complement traditional output-based indicators by capturing the demand-side dimensions of economic activity. Third, this article incorporates Comprehensive Wealth per capita and Human Capital per capita. Recognizing that GDP fails to capture certain forms of productive activity—particularly those derived from natural capital—Comprehensive Wealth provides a broader measure of national wealth, reflecting the full asset portfolio of a country and its long-term development trajectory. Simultaneously, given the increasing significance of labor force participation and returns to education for long-term economic sustainability, Human Capital should be included as a critical dimension of development. Finally, three indirect indicators—nighttime lights, electricity production, and rail line density—are employed to proxy economic development through infrastructure and investment capacity. These variables serve as credible proxies for the level of economic activity, as higher infrastructure provision is often indicative of a more advanced stage of economic development. Due to space limitations in the main text, the subsequent empirical analyses present only the regression results using GDP per capita growth (GDPpc_growth) and GNI per capita growth (GNIpc_growth) as the explained variables. Full regression results for the other explained variables are detailed in the [Supporting Information](#).

Additionally, this article also introduces 12 control variables (corresponding to x), including: government intervention (government), natural resource (resource), capital formation (capital), industry production (industry), urbanization (urbanize), Employment in Industry (EI), Openness of market, employment rate (employment), Foreign Direct Investment (FDI), international reserve (reserve), trade with China (tiva), technology cooperation with China (coop). All abbreviations of variables with their corresponding full names and measurements, used in baseline regressions, are detailed in Table S3 (in [Supporting Information](#)).

The descriptive statistics of each variable are shown in Table 5. It is worth mentioning that the sample data used in the baseline regression includes only the data from 1990 to 2021 of 41 “Belt and Road” countries used to measure GPYCs in the previous text. Additionally, the sample data for each variable has been winsorized by 1% (at both tails).

4.2 | Endogeneity Handling and Estimation Approach

Actually, it is difficult to realize comprehensiveness in selecting effective control variables within empirical models of economic growth, which means that there may be endogeneity issues caused by variable omission. In addition, there may be a reverse causation between knowledge complexity and economic growth. Knowledge complexity should have a positive impact on economic growth, but the latter may further promote the development of related industries, thus leading to a further increase in the former. To eliminate such endogeneity, this article introduces instrument variables to conduct two-stage regression and, on this basis, introduces both

TABLE 5 | Descriptive statistics of each variable.

Statistic	N	Mean	St. dev.	Min	Max
GDPpc_growth	1312	0.027	0.047	−0.086	0.149
GNIpc_growth	1312	0.024	0.041	−0.086	0.148
Household	1197	0.041	0.071	−0.086	0.432
CO ₂	1184	1.678	0.911	−0.086	3.465
DCWpc	965	0.062	0.122	−0.086	0.489
HCpc	975	0.076	0.128	−0.086	0.504
Light	1150	0.122	0.466	−0.086	4.728
Elec	1312	7.871	0.920	5.078	9.797
Rail	853	8.412	1.317	5.613	11.606
GPYC_rank	1312	0.489	0.260	0.010	1.000
LP	902	0.053	0.095	0.000	0.286
LO	1312	15.194	0.410	14.230	15.487
Government	1237	0.773	0.144	0.430	1.171
Resource	1123	0.035	0.049	0.0003	0.310
Capital	1239	0.256	0.070	0.053	0.466
Industry	1201	0.304	0.096	0.064	0.598
Urbanize	1312	0.012	0.019	−0.030	0.072
EI	1271	0.254	0.071	0.082	0.420
Open	1249	0.766	0.478	0.131	3.308
Employment	1271	0.535	0.099	0.294	0.792
FDI	1241	0.052	0.162	−0.086	2.793
Reserve	1179	0.103	0.114	0.004	0.669
Tiva	780	0.011	0.010	0.000	0.054
Coop	1230	2.681	2.475	0.000	9.508

time and individual fixed effects into the regression to eliminate endogeneity as much as possible.

This article considers using the geographic factor as one of the instrumental variables, which can be seen to be absolutely exogenous (Hall and Jones 1999). Specifically, this article selects the average geographic distance between “Belt and Road” countries and the US. This variable will be effective: Given higher levels of economic development, technology, and education in the US, it is more inclined to go to the US for further studies and obtain technology spillovers. Therefore, a farther geographical distance between a country and the US means a higher cost of studying, which hinders technology spillovers. Moreover, this article considers using the language factor as an instrumental variable. Specifically, a language dummy variable will be used to indicate whether the mainstream language of a country is English; it is worth noting that the mainstream language refers to the first or second official language, or the most widely spoken language. This variable should also be effective: language factors are almost exogenous (Mauro 1995); and the writing language required

by international core journals is English, so countries where English is the mainstream language are more likely to acquire cutting-edge knowledge.

However, it is worth noting that the geographic distance and language variables are not time-varying, which means that they will be excluded by the introduction of individual fixed effects. Therefore, this article will multiply the geographic distance variable with the educational expenditure per student in the US and the language variable with the articles published in core journals by the US, and then interprets the resulting indexes as “Learning Opportunity (LO)” and “Learning Possibility (LP)”, respectively.

In most empirical analyses, linear regression based on OLS examines the “average effect” of explanatory variables on the explained variable. However, it also means that regression results will often be influenced by outliers, and such an issue will be particularly pronounced while performing grouped regressions. More importantly, when the conditional distribution $y|x$ does not meet the criteria for symmetric distribution, regression results based on $E(y|x)$ can hardly depict the full landscape of the conditional distribution. Therefore, this article considers using Quantile Regression (QR) as the main estimation method for empirical analyses, because QR has strong robustness against outliers, nonnormality of residuals, and heteroscedasticity issues. Additionally, QR allows us to investigate heterogeneity across different quantiles in the regression directly.

It is worth mentioning that to further incorporate settlements for endogeneity and fixed effects into the econometric exercises; we consider adopting the Instrumental Variable Fixed Effect Quantile Regression (IV-FEQR) as the main estimation method. Consider a multivariate regression model $\mathbf{Y} = \beta\mathbf{X} + \varepsilon$, the Loss Function referring to QR can be defined as: a . And then, the estimator $\hat{\beta}$ of β corresponds to the solution of:

$$\begin{aligned} \hat{\beta}(\tau) &= \arg \min_{\beta \in R_p} \sum_{i=1}^n \rho_{\tau}(y_i - \beta'(\tau)\mathbf{x}_i) \\ &= \arg \min_{\beta} \sum_{y_i \geq \beta'(\tau)\mathbf{x}_i} \tau(y_i - \beta'(\tau)\mathbf{x}_i) + \sum_{y_i < \beta'(\tau)\mathbf{x}_i} (\tau-1)(y_i - \beta'(\tau)\mathbf{x}_i) \end{aligned} \quad (7)$$

Given that reverse causation may result in endogeneity, this article adopts the QR based on Instrumental Variable (IV-QR) introduced by Abadie et al. (2002), Chernozhukov and Hansen (2004, 2008) and Harding and Lamarche (2009). Consider a two-stage model:

$$\begin{cases} \mathbf{Y} = \Theta\mathbf{X} + \mathbf{u} \\ \mathbf{X} = \Pi\mathbf{Z} + \mathbf{v} \end{cases} \quad (8)$$

Within two-stage IV-QR, the estimator $\hat{\Theta}$ of Θ will correspond to the solution of minimizing the Loss Function $R(\Theta)$:

$$\begin{aligned} \hat{\Theta} &= \arg \min_{\Theta} R(\Theta) = \arg \min_{\Theta} \sum_{i=1}^n \rho_{\tau}(y_i - \Theta'(\tau)\mathbf{x}_i) \\ &= \min_{\Theta} \left[Y_{it}\Theta(\tau)\tilde{\mathbf{X}}_{it} \sum \tau(Y_{it} - \Theta'(\tau)\tilde{\mathbf{X}}_{it}) + Y_{it}\Theta(\tau)\tilde{\mathbf{X}}_{it} \sum (\tau-1)(Y_{it} - \Theta'(\tau)\tilde{\mathbf{X}}_{it}) \right] \end{aligned} \quad (9)$$

where $\tilde{\mathbf{X}}_{it}$ denotes the fitted values of endogenous variable(s) in the first stage, which can be calculated by:

$$\begin{cases} \tilde{\mathbf{X}} = \mathbf{Z}\hat{\Pi} \\ \hat{\Pi} = (\mathbf{Z}^T\mathbf{Z})^{-1}\mathbf{Z}^T\mathbf{X} \end{cases} \quad (10)$$

In order to introduce fixed effects based on IV-QR, this article adopts a two-stage regression proposed by Canay (2011). Consider a model including time and individual fixed effects:

$$Y_{it} = \theta' \mathbf{X}_{it} + \lambda_t + \mu_i + \varepsilon_{it}, \quad E(\varepsilon_{it} | \mathbf{X}_{it}, \lambda_t, \mu_i) = 0 \quad (11)$$

Let $\hat{\theta}$ be the first-stage estimator of θ , and define $\hat{\lambda}_t = \mathbb{E}_n[Y_{it} - \theta' \mathbf{X}_{it}] = \sum_{i=1}^n \rho_{\tau}(Y_{it} - \hat{\theta}' \mathbf{X}_{it})/n$, $\hat{\mu}_i = \mathbb{E}_T[Y_{it} - \theta' \mathbf{X}_{it}] = \sum_{t=1}^T \rho_{\tau}(Y_{it} - \hat{\theta}' \mathbf{X}_{it})/T$. Then let $\hat{Y}_{it} \equiv Y_{it} - \hat{\lambda}_t - \hat{\mu}_i + \bar{Y}_{it}$, and the second-stage estimator $\hat{\theta}(\tau)$ can be written as:

$$\hat{\theta}(\tau) = \arg \min_{\theta} \mathbb{E}_{nT}[\hat{Y}_{it} - \theta' \mathbf{X}_{it}] = \sum_{i=1}^n \sum_{t=1}^T \rho_{\tau}(\hat{Y}_{it} - \hat{\theta}' \mathbf{X}_{it})/nT \quad (12)$$

5 | Empirical Evidence Within the “Belt and Road” Initiative

5.1 | Baseline Result With Robustness Check

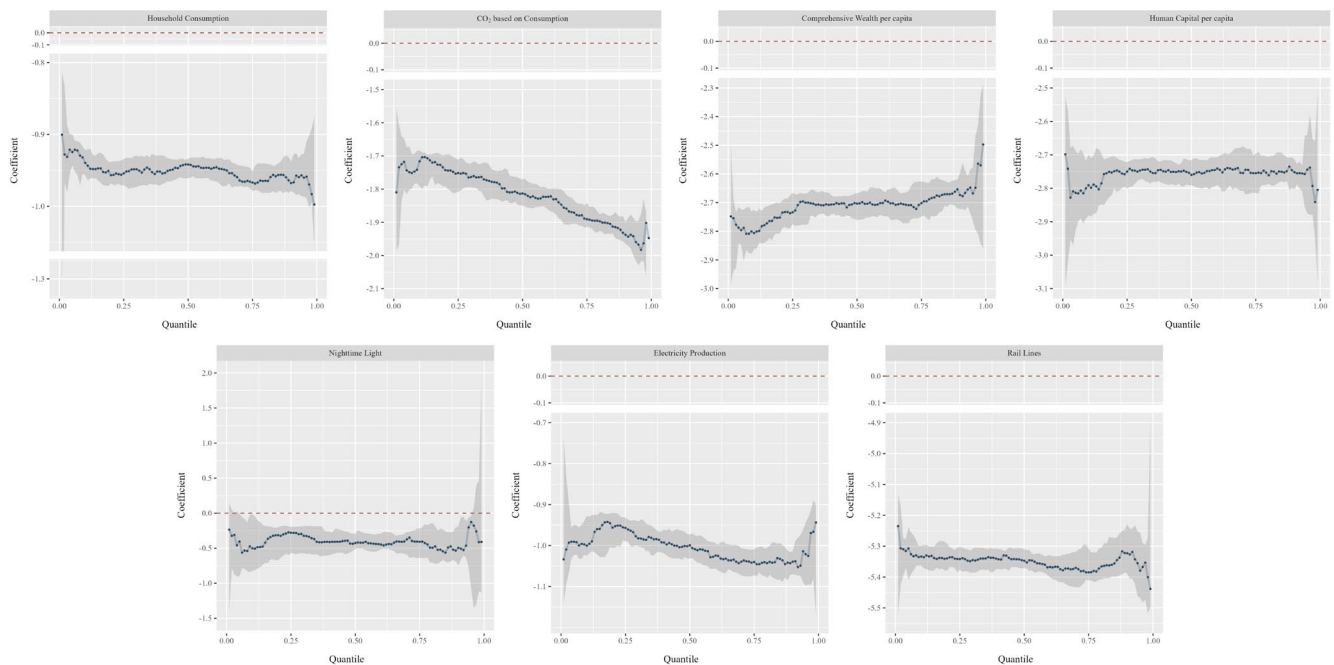
Based on Equation (6) and Equation (12), this article conducts regressions under different quantiles. Due to space limitations in the main text, Table 6 reports selected coefficients of GPYC_rank for the 0.2–0.8 quantiles (seven quantile points in total) with GDPpc_growth and GNipc_growth as the explained variables. The full results of all coefficients are available in Tables S8 and S9, while Figure S1 displays the full regression results across all quantiles from 0.01 to 0.99. Additionally, Figure 4 presents the regression results for the remaining seven explained variables across all quantiles, ranging from 0.01 to 0.99.

The baseline regression results are in line with our expectations. The results in Table 6 show that, the coefficients of GPYC_rank are all significantly negative at the 1% level under different quantiles, which means that the increase of knowledge complexity will indeed have a positive impact on promoting economic growth; and actually, the coefficients suggest that the impact of knowledge complexity on economic growth should be quite considerable—Every 1% point decrease in GPYC_rank (i.e., an upward shift in the ranking place of GPYC) is associated with an approximate 0.9% point increase in GDPpc_growth. From the view of the absolute values of GPYC_rank coefficients, there is no obvious difference across different quantiles, which implies that although there may be differences in the economic growth rates of different countries, the promoting effect of knowledge complexity on economic growth is generally consistent. Results based on the other seven dependent variables indicate considerable robustness: the coefficients of GPYC_rank remain significantly negative at sufficiently high levels and are almost consistent

TABLE 6 | Results of baseline regressions with GDPpc_growth as explained variable.

Variable	GDPpc_growth						
	(1) $\tau=0.2$	(2) $\tau=0.3$	(3) $\tau=0.4$	(4) $\tau=0.5$	(5) $\tau=0.6$	(6) $\tau=0.7$	(7) $\tau=0.8$
<i>Panel A: GDPpc_growth ~</i>							
GPYC_rank	−0.891***	−0.886***	−0.891***	−0.897***	−0.899***	−0.899***	−0.895***
Control variable	✓	✓	✓	✓	✓	✓	✓
Observation	596	596	596	596	596	596	596
Year FE	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓
R_1	0.901	0.902	0.903	0.902	0.901	0.901	0.901
<i>Panel B: GNIpc_growth ~</i>							
GPYC_rank	−0.586***	−0.596***	−0.594***	−0.596***	−0.593***	−0.593***	−0.591***
Control variable	✓	✓	✓	✓	✓	✓	✓
Observation	596	596	596	596	596	596	596
Year FE	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓
R_1	0.843	0.843	0.845	0.845	0.844	0.846	0.848

Note: *** denote $p < 0.01$.

**FIGURE 4** | Results of regressions based on different aspects of Economic Development. Each sub-figure shows the results of GPYC_rank coefficients under 0.01–0.99 quantiles. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/rdev.12056)]

across almost all quantiles.³ Moreover, from the perspective of the two-stage model itself, the instrument variables are also effective sufficiently, the results of first stage of IV-FEQR with their diagnoses are detailed in Table S7. Therefore, it is believable that the baseline regression results align with our expectations for Hypothesis 1.

To verify the robustness of the baseline findings, a series of robustness checks were conducted, including the following procedures:

- Core explanatory variable replacement. This article replaces GPYC_rank with GPYC (the Max–Min standardized value of the original GPYC index) and re-conducts regressions.

- Anticipation effect. To test whether expectations of economic growth may be influenced by technological investment and patenting behavior, this article employed the lead 1 term of GPYC_rank as the core explanatory variable.
- Time-lag effect. Given that improvements in knowledge complexity may manifest in time continuity, this article incorporates lagged 1, 2, and 3 terms of GPYC_rank into regressions to investigate such a time-lag effect.
- Additional control variables. To address potential omitted variable bias arising from an incomplete set of covariates, six additional control variables were included: Total Factor Productivity (TFP), education expenditure (education), school enrollment (enroll), government effectiveness (gov. eff.), law and order (law.order), and government accountability (acct.). The abbreviations, definitions, measurement methods, and descriptive statistics for these variables are reported in Tables S4 and S5.
- Period adjustment. The baseline panel dataset spans from 1990 to 2021. However, given the evolving nature of global patenting and the occurrence of major global shocks during this period, this article conducts two period adjustments. First, to remove potential policy effects associated with the formal launch of the Belt and Road Initiative in March 2015, we excluded observations from the period after 2015. Second, to address disruptions caused by the 2007–2008 global financial crisis, this article excludes observations

prior to 2009. Regressions are re-conducted based on two adjusted samples.

- Country adjustment. The panel dataset includes 41 BRI countries. Given China's central role in initiating the BRI and its comparatively strong economic and diplomatic influence, unobserved heterogeneity may result in biases in the results. Therefore, this article adopts four exclusion strategies to isolate potential confounders: First, countries with high-level diplomatic or strategic partnerships with China are excluded.⁴ Second, this article excludes countries that have established a bilateral “Community of Shared Destiny” with China.⁵ Third, considering that the Belt and Road Initiative was officially launched in 2015, this article excludes initial countries that participated in the BRI, which are defined as those that signed a Memorandum of Understanding (MoU) or a cooperation agreement with China prior to 2015.⁶ Finally, this article excludes the 37 regional prospective founding members of the Asian Infrastructure Investment Bank (AIIB), which was officially established in 2015.⁷ Subsequent regressions were re-conducted based on the adjusted samples.

Robustness check results are summarized in Figure 5, which displays results at the 0.25, 0.50, and 0.75 quantiles using GDPpc_growth and GNIpc_growth as explained variables. Corresponding results for the other seven explained variables are provided in Table S14 through Table S20.

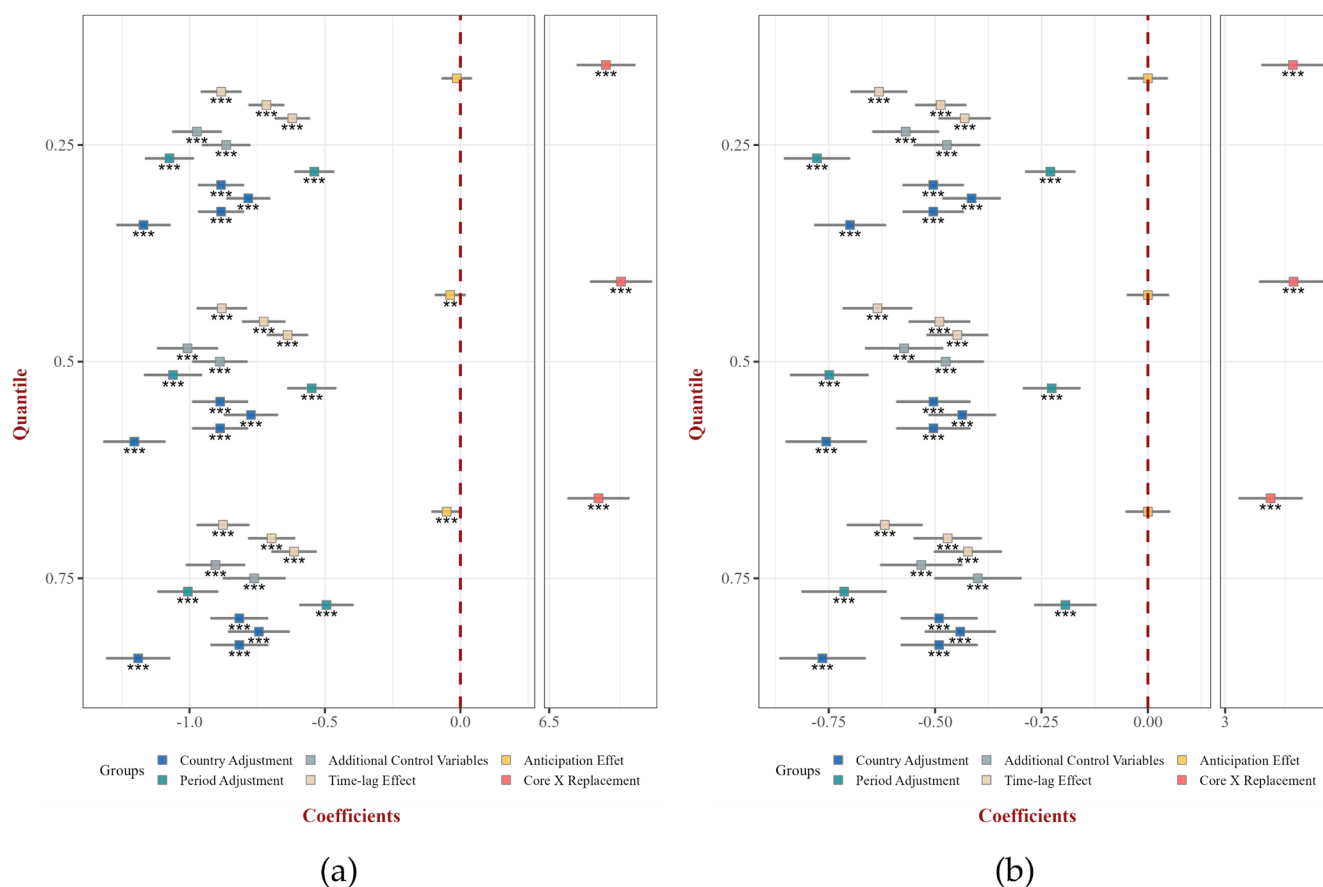


FIGURE 5 | Results of robustness checks. Subfigure (a) shows results of robustness checks with GDPpc_growth as the explained variable; Subfigure (b) shows the results of GNIpc_growth. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

The results of the robustness checks largely support the findings of the baseline regressions. The significantly positive coefficients of GPYC reinforce the main argument that knowledge complexity enhances economic growth. While no substantial evidence was found for an anticipation effect of patent output on economic growth, there is clear support for a time-lag effect, for example, the positive impact of knowledge complexity persists over time but tends to attenuate as the lag increases. Furthermore, the inclusion of additional control variables does not materially alter the coefficients, indicating that the baseline results are not driven by omitted variable bias. While the sample is restricted to exclude post-2015 observations or pre-2009 observations, the results remain broadly consistent with the baseline regressions. It implies that neither the long temporal span of the data nor major global economic shocks significantly biases the core findings. Last, a series of country-sample adjustments still yielded robust results. It suggests that bilateral diplomatic intensity with China is not a confounding factor in explaining the effect of knowledge complexity on economic growth in BRI countries.

5.2 | Mechanism Tests

As hypothesized earlier, this article posits that knowledge complexity promotes economic growth through both macro-level and micro-level mechanisms. At the macroeconomic level, increases in knowledge complexity strengthen the international competitiveness of countries by facilitating high-technology patent output and the export of advanced technological products. Simultaneously, higher knowledge complexity promotes industrial development by enhancing inter-industry linkages and facilitating the optimization of industrial structure. At the microeconomic level, knowledge complexity contributes to economic growth by enabling learning-by-doing in innovation processes, thereby stimulating the production of high-value-added goods. Furthermore, complex knowledge tends to accumulate over time, generating sustained technological progress and continuous value creation. To empirically test these two mechanisms, this article conducts grouped regressions based on mediating variables. Specifically, this article examines whether the coefficients of GPYC_rank differ significantly across subgroups defined by the mediating variables, thereby shedding light on the channels through which knowledge complexity affects economic growth.

First, for the macro-level mechanisms, this article uses (1) intellectual property charges and (2) value-added in the secondary and tertiary industries as two mediating variables. (corresponding to charge and optimize, respectively). Actually, the R&D capability of products with higher technological content can be measured by the output of high-tech patents, and the economic value contained in such products can be measured by the charges for the use of intellectual property. The higher the charges received by countries, the more high-tech and valuable products they can produce. On the other hand, the optimization of industrial structure generally means that the dominant industry of a country shifts from low-value-added labor-intensive industries to high-value-added technology-intensive industries. Therefore, the optimization of industrial structure can be measured by the added value in the field of medium- and high-end manufacturing and service sectors.

For the micro-level mechanisms, this article uses (3) the scale of technicians in R&D and (4) expenditures on R&D as two mediating variables (corresponding to technicians and R&D_expenditure, respectively). Enterprises that participate in international markets and engage in high-tech production often accumulate new knowledge through extensive business practices. In fact, the participation of enterprises in high-tech business practices usually requires the involvement of technicians or researchers. A larger group of technicians typically means higher levels of R&D practice, leading to higher levels of complex knowledge accumulation. On the other hand, creating higher innovation pressure can usually stimulate enterprises to carry out faster technological innovations and achieve higher levels of technological accumulation. Facing higher innovation pressure, enterprises usually choose to increase their R&D expenditures in order to maintain their position in the industry and maintain their competitive advantages, thereby achieving further technological innovation.

For each of the four mediators, the sample of BRI countries used in baseline regressions is partitioned into high and low groups based on the median value. Grouped IV-FEQR estimations are conducted for each subgroup. Columns (1) through (4) in Table 7 report the results at the median quantile; results for other quantiles are available in Table S21 through Table S24.

Results show that, within samples based on four partition criteria, there are significant differences in GPYC_rank coefficients between the two sub-samples. The results are consistent with our expectations for the four sub-hypotheses in Hypotheses 2a, 2b, 3a and 3b, namely, the macroeconomic approach, incorporated with the microeconomic one, of knowledge complexity's promoting economic growth can be proven.

5.3 | Heterogeneity Analyses

Previous studies have shown that complex knowledge exhibits “spatial stickiness” (Balland and Rigby 2017; Sorenson et al. 2006), which hinders its diffusion among regions, leading to certain locational differences and forming certain geographical patterns (Frenken and Boschma 2007; Lawson and Lorenz 1999). In order to investigate the regional heterogeneity of the impact of knowledge complexity on economic growth, this article partitions the sample based on geographical or ethnocultural region to which the countries belong, including the ASEAN (Association of Southeast Asian Nations), Europe, CIS (Commonwealth of Independent States), South Asia and West Asia, and then conducts IV-FEQR. Columns (1) through (5) in Table 8 report the results at the median quantile; results for other quantiles are available in Table S25.

The results show that, as for European countries, the coefficients of GPYC_rank are significantly positive at the 1% level, but for ASEAN, CIS, South Asian, and West Asian countries, the coefficients of GPYC_rank are all significantly negative at the 1% level. Thereinto, the ASEAN group exhibits the smallest absolute coefficients, while the CIS group shows the largest. Additionally, there is no overlap of 97.5% confidence intervals among the groups, which means that the positive

TABLE 7 | Results of mechanism tests.

Variable	Mechanism test			
	Property charge	Industrial optimization	R&D practice activity	R&D expenditure
	(1)	(2)	(3)	(4)
<i>Panel A: (Total) mediator ~</i>				
GPYC_rank	−0.451*** (0.035)	−0.511*** (0.003)	−1.291*** (0.019)	−5.670*** (0.020)
Control variable	✓	✓	✓	✓
Observation	1502	1780	810	1481
Year FE	✓	✓	✓	✓
Country FE	✓	✓	✓	✓
R_1	0.421	0.812	0.700	0.874
<i>Panel B: (Low mediator) GDPpc_growth ~</i>				
GPYC_rank	−0.270*** (0.004)	−0.646*** (0.004)	−0.190*** (0.006)	−0.270*** (0.005)
Control variable	✓	✓	✓	✓
Observation	673	867	401	669
Year FE	✓	✓	✓	✓
Country FE	✓	✓	✓	✓
R_1	0.714	0.861	0.659	0.698
<i>Panel C: (High mediator) GDPpc_growth ~</i>				
GPYC_rank	0.047*** (0.003)	0.091*** (0.003)	0.105*** (0.004)	0.111*** (0.003)
Control variable	✓	✓	✓	✓
Observation	829	913	409	812
Year FE	✓	✓	✓	✓
Country FE	✓	✓	✓	✓
R_1	0.579	0.670	0.678	0.683

Note: *** denote $p < 0.01$.

TABLE 8 | Results of samples grouped by geographical or ethno-cultural region.

Variable	Economic growth				
	ASEAN	West Asia	South Asia	CIS	Europe
	(1)	(2)	(3)	(4)	(5)
GPYC_rank	−0.472*** (0.013)	−0.774*** (0.011)	−2.427*** (0.044)	−7.522*** (0.028)	1.672*** (0.010)
Control variable	✓	✓	✓	✓	✓
Observation	128	201	56	116	261
Year FE	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓
R_1	0.880	0.868	0.989	0.986	0.917

Note: *** denote $p < 0.01$.

impact of knowledge complexity on economic growth has significant regional heterogeneity. In fact, the regional heterogeneity implies the macroeconomic approach to some extent. There are a considerable number of countries that can be regarded as having a high level of technological development in Europe and the ASEAN, such as Japan, Singapore, Germany, etc. In contrast, most countries in South Asia, West Asia, and the CIS have relatively limited technological innovation capabilities. Therefore, it can be confirmed that the improvement of knowledge complexity can promote economic growth by enhancing international competitiveness and optimizing industrial structure.

This article presumed the long-term “validity” of patents while measuring knowledge complexity in the previous text. However, in reality, a patent usually loses its exclusivity after a certain period of time from the date of application, allowing widespread acquisition of the knowledge it once covered. Alternatively, a patent may initially cover highly complex knowledge in a certain domain, but as the domain becomes more accessible to outsiders, the previously highly complex knowledge becomes widely understood. In both cases, the complex knowledge covered by the patent will no longer be considered complex, and the patent can be seen as no longer contributing to the knowledge complexity of the corresponding domain.

Therefore, this article considers setting “expirations” of 15, 10, and 5 years, and then re-calculates GPYCs based on re-constructing the bipartite network by defining $P_{ct,y} = \sum_{i=y-exp}^y N_{ct,i}$ mentioned above. The results at the median quantile are shown in Columns (1) through (4) in Table 9; results for other quantiles are available in Table S26.

Across all groups, the coefficients of GPYC_rank remain significantly negative at sufficiently high levels. Notably, as the statistical interval for patent output shortens, the absolute value of the GPYC_rank coefficient increases. This implies that the promoting effect of knowledge complexity is stronger in the context of more recent, advancing innovation. This finding has meaningful implications, which suggest that the momentum of economic growth increasingly depends on the ongoing production

of cutting-edge technologies, rather than on inherited or previously accumulated technological advantages. A country capable of consistently generating high-technology patents is more likely to sustain economic dynamism (for its outstanding performance in patenting, even though the statistical interval is short). In contrast, countries that rely heavily on legacy technologies without renewing their innovation capacity may see the growth benefits of past technological leadership gradually diminish.

As noted earlier, in calculating knowledge complexity based on patent data across 35 distinct technological domains, this article accounts for both the “heterogeneity” of complex knowledge across different domains and their interconnectedness. This design substantially mitigates the risk of overestimating the contribution of technologically “simple” domains while avoiding the underestimation of genuinely “complex” ones. Nonetheless, the construction of a comprehensive index inevitably compresses the original data, making it difficult to assess the heterogeneous effects of individual technological domains on economic growth.

To address this limitation, this article investigates whether the impact of complex knowledge production on economic growth differs significantly across the 35 technological domains. Since the knowledge complexity indicators are built upon a bipartite network structure using the Revealed Comparative Advantage (RCA) of countries in each technological domain, based on patent output data, this article estimates separate regressions using the RCA index of each domain as the core explanatory variable. Given that a higher RCA implies more advanced or specialized technological output, this article expects domains with more complex knowledge production to exhibit a stronger positive effect on economic growth, reflected in higher positive estimated coefficients.

On the other hand, while this article employs nine distinct indicators to measure economic development both directly and indirectly, the utilization of composite measures likewise risks masking sector-specific heterogeneity, especially given the structural peculiarities across industries within national economies. The OECD ICIO table provides detailed data on international economic linkages across 45 sectors in 77 countries or regions, including domestic output for each sector. Thus, this article also

TABLE 9 | Results of samples grouped by “expirations” of patents.

Variable	Economic growth			
	Long-term (1)	15 years (2)	10 years (3)	5 years (4)
GPYC_rank	−0.897*** (0.005)	−0.975*** (0.005)	−1.328*** (0.005)	−1.532*** (0.006)
Control variable	✓	✓	✓	✓
Observation	596	487	488	484
Year FE	✓	✓	✓	✓
Country FE	✓	✓	✓	✓
R_1	0.902	0.904	0.928	0.937

Note: *** denote $p < 0.01$.

estimates the effects of knowledge complexity on output growth across 45 different economic sectors (specifications of all 45 sectors are provided in Table S27).

Although the classification schemes of technological domains and economic sectors are not directly compatible, which prevents the construction of a fully integrated panel dataset, this article also conducts sensitivity analyses of how complex knowledge from different technological domains influences various economic sectors. Specifically, we examine the heterogeneous effects of the 35 technological fields (as captured by RCA indices) on the domestic output of 45 sectors using the OECD ICIO dataset.

The results of these three heterogeneity analyses are presented in Figure 6. First, (a) shows that not all technological domains contribute positively to economic growth. However, the domains with positive RCA coefficients⁸ largely overlap with those previously identified as high in knowledge complexity, while those with negative coefficients⁹ are also mostly low-complexity domains—results that are broadly consistent with our expectations. Second, (b) indicates that knowledge complexity exerts a significant promoting effect on the output of most sectors. The main heterogeneity lies in the relatively weaker effects observed in labor-intensive, low-knowledge industries such as animal husbandry, mining, and low-end manufacturing.¹⁰ In contrast, industries with higher value-added show more consistent and robust gains from increased knowledge complexity. Lastly, the results from the sensitivity analyses of the effects between technological domains and economic sectors are broadly aligned with the above findings, reinforcing the robustness of our conclusions regarding the heterogeneous impact of complex knowledge across domains and sectors.

6 | Whether the “Belt and Road” Differ?

6.1 | Comparing Globally

Since the “Belt and Road” Initiative was proposed, the countries along the “Belt and Road” indeed experienced rapid development, which speaks for themselves. Moreover, considerable research indicates that such a multilateral cooperation mechanism as the “Belt and Road” can promote infrastructure construction and the flow of resources, thereby exerting a significant impact on regional development (Erbach 2021; Huang 2016; Wang et al. 2020; Wang 2019). Therefore, we aim to examine whether the improvement in economic growth due to knowledge complexity in “Belt and Road” countries is significantly different from that in other countries, that is, whether the “Belt and Road” has indeed played an indispensable role.

Comparing confidence intervals is a conventional method. Therefore, this article first sets up a “global sample” composed of the 100 countries used in measuring GPYCs in the previous text, including 41 “Belt and Road” countries and 59 “non-Belt and Road” countries (descriptive statistics are shown in Table S6). Then, this article conducts IV-FEQR based on the “global sample” (results are shown in Table 10) and calculates the 97.5% confidence intervals for the “Belt and Road” sample and the “global” sample (results are shown in Table S28).

The results show that based on the global sample, the coefficients of GPYC_rank are significantly negative at the 1% significance level under different percentiles. This means that for most countries worldwide, improving knowledge complexity can indeed provide impetus for improving their economic growth. Moreover, the absolute values of the coefficient of GPYC_rank

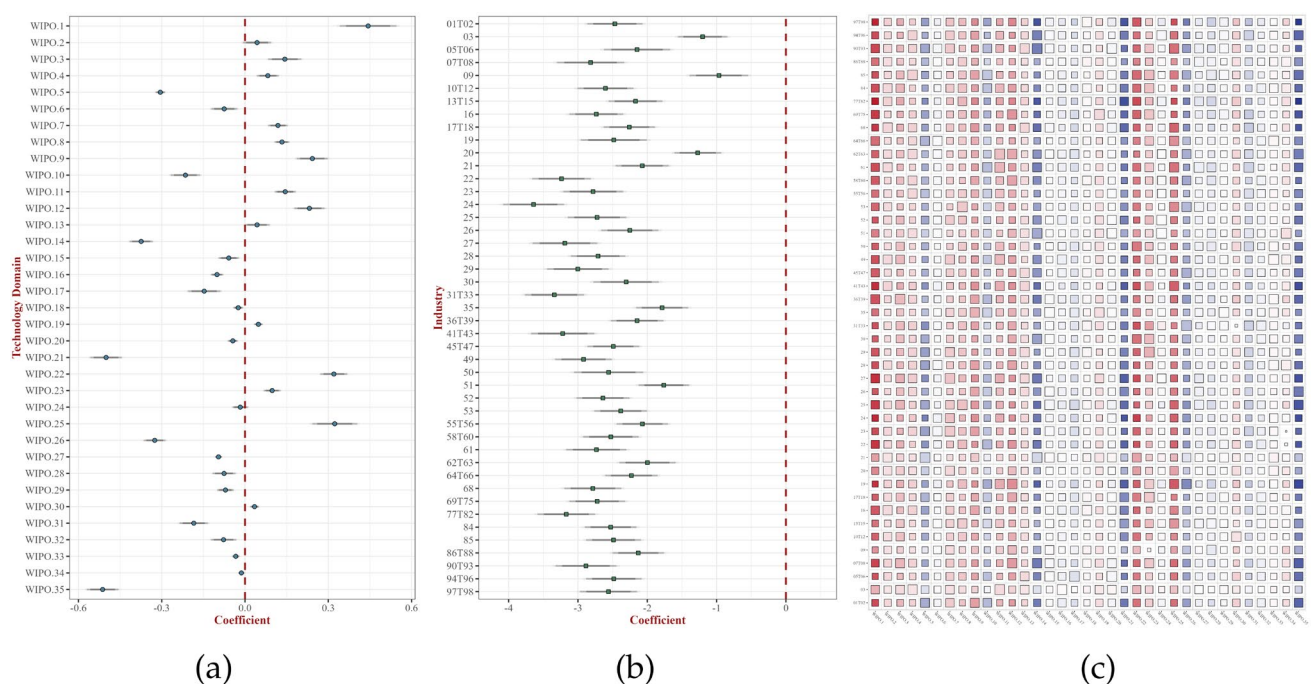


FIGURE 6 | Results of heterogeneous analyses and sensitivity analyses. Subfigure (a) shows results of heterogeneous analyses based on technology domain division; Subfigure (b) shows the results based on industry division; Subfigure (c) shows the results of sensitivity analyses. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

based on the “Belt and Road” sample are higher than those in the “global sample” significantly. This indicates that the “Belt and Road” Initiative has provided participating countries with a better environment for economic development and the creation and flow of knowledge factors, which enables the improvement of knowledge complexity to promote economic growth from a higher level. Thus, it can be considered that Hypothesis 4 presented above has been considerably validated.

Further, in order to evaluate the “effect” of the “Belt and Road” on the relationship between knowledge complexity and economic growth, and eliminate endogeneity caused by self-selection bias, this article considers combining Difference-in-Difference (DID) with the moderating effect model, incorporating with Propensity Score Matching (PSM),¹¹ and still conducts estimation based on IV-FEQR. In order to test the moderating effect in the DID model, this article introduces $GPYCr_{it}$ and interaction term $GPYCr_{it} \times if_B\&R_{it} = GPYCr_{it} \times treat_i \times post_t$ to reflect the knowledge complexity of the participating and nonparticipating countries:

$$GDPpc_growth_{it} = \alpha + \beta GPYCr_{it} + \gamma GPYCr_{it} + \delta GPYCr_{it} \times if_B\&R_{it} + \sum_{j=2}^J \zeta_j x_{j,it} + \lambda_t + \mu_i + \varepsilon_{it} \quad (13)$$

where λ_t denotes time-fixed effects; μ_i denotes individual-fixed effects; δ^{did} is the focus within the DID-moderation model. It is worth noting that the “DID” mentioned in this article is merely regarded as part of the moderation effect test, still equivalent to the TWFE estimation of DID, rather than the narrowly defined one used to evaluate policy effects. In fact, we only focus on the coefficient of the interaction term between the core explanatory variable and the moderator in the moderation model to determine whether there is a significant moderating effect, rather than evaluating the net policy effect of the “Belt and Road.” Therefore, it is unnecessary to discuss the series of strict assumptions related to the DID model; and the “Belt and Road” can be regarded as a fundamentally exogenous policy shock, which is sufficient to ensure the good estimation properties of the interaction term coefficients. We only conducted a parallel trend test here, as shown in Figure 7;¹² the test results indicate that the parallel trend assumption for the treatment and control groups before the policy implementation year cannot be rejected.

The regression results are shown in Table 11. Results show that almost all the coefficients δ^{did} of the interaction term are negative at the 1% significance level under different quantiles. Thus, it can be concluded that, as we expected, compared to countries not participating in the “Belt and Road”, the knowledge complexity of “Belt and Road” countries significantly enhance economic

TABLE 10 | Results of “global sample.”

Variable	GDPpc_growth						
	(1) $\tau=0.2$	(2) $\tau=0.3$	(3) $\tau=0.4$	(4) $\tau=0.5$	(5) $\tau=0.6$	(6) $\tau=0.7$	(7) $\tau=0.8$
GPYC_rank	−0.038*** (0.003)	−0.035*** (0.003)	−0.032*** (0.003)	−0.031*** (0.003)	−0.032*** (0.003)	−0.030*** (0.002)	−0.030*** (0.004)
Observation	1142	1142	1142	1142	1142	1142	1142
Time FEs	✓	✓	✓	✓	✓	✓	✓
Individual FEs	✓	✓	✓	✓	✓	✓	✓
Pseudo R^2_1	0.414	0.430	0.435	0.436	0.436	0.434	0.422

Note: *** denote $p < 0.01$.

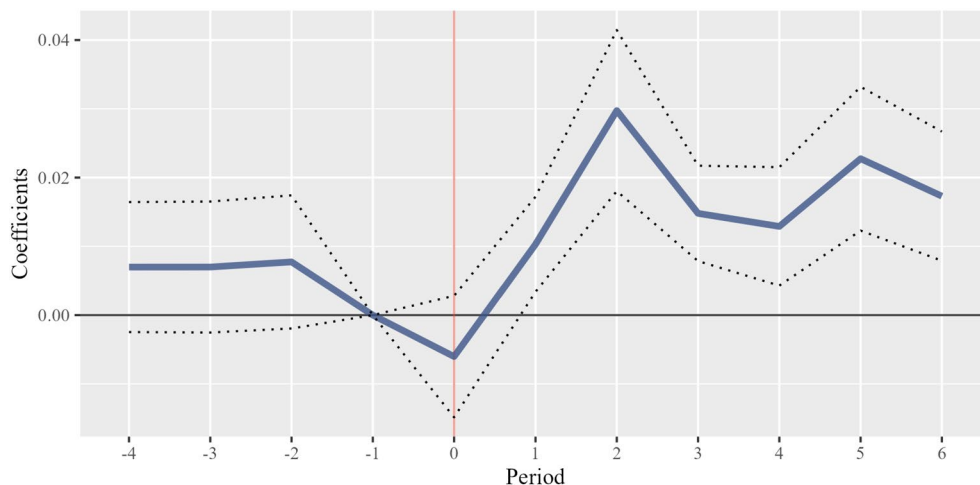


FIGURE 7 | Results of parallel trend test. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

growth levels to a greater extent; namely, the “Belt and Road” will exert a significant positive effect on knowledge complexity’s promoting economic growth. Additionally, we conducted placebo tests based on 100 bootstraps for the aforementioned results, as shown in Figure 8. The results indicate that the moderating effect coefficient significantly deviates from the estimation result clusters obtained through IVFEQR based on random sampling, confirming the reliability of the regression results in this section.

6.2 | How Will External Factors Interact?

In the preceding analysis, this article has demonstrated that, under the framework of the Belt and Road Initiative, the production of complex knowledge contributes to economic growth through channels of industrial upgrading and innovation-driven development. Moreover, it has emphasized the significant positive moderating effect of the BRI in enhancing such a promoting effect. However, against the backdrop of increasing geopolitical

TABLE 11 | Moderating effect results of Belt and Road.

Variable	Economic growth						
	(1) $\tau=0.2$	(2) $\tau=0.3$	(3) $\tau=0.4$	(4) $\tau=0.5$	(5) $\tau=0.6$	(6) $\tau=0.7$	(7) $\tau=0.8$
GPYC_rank	−0.038*** (0.006)	−0.036*** (0.006)	−0.035*** (0.005)	−0.032*** (0.005)	−0.031*** (0.004)	−0.032*** (0.004)	−0.034*** (0.006)
BRI	−0.004 (0.004)	−0.003 (0.004)	−0.002 (0.003)	0.003 (0.003)	0.005* (0.003)	0.007*** (0.003)	0.009* (0.005)
GPYC_rank × BRI	−0.104*** (0.011)	−0.103*** (0.008)	−0.104*** (0.007)	−0.113*** (0.007)	−0.116*** (0.006)	−0.118*** (0.005)	−0.118*** (0.011)
Control variable	✓	✓	✓	✓	✓	✓	✓
Observation	657	657	657	657	657	657	657
Year FE	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓
R_1	0.645	0.660	0.667	0.673	0.678	0.687	0.693

Note: *** denote $p < 0.01$.

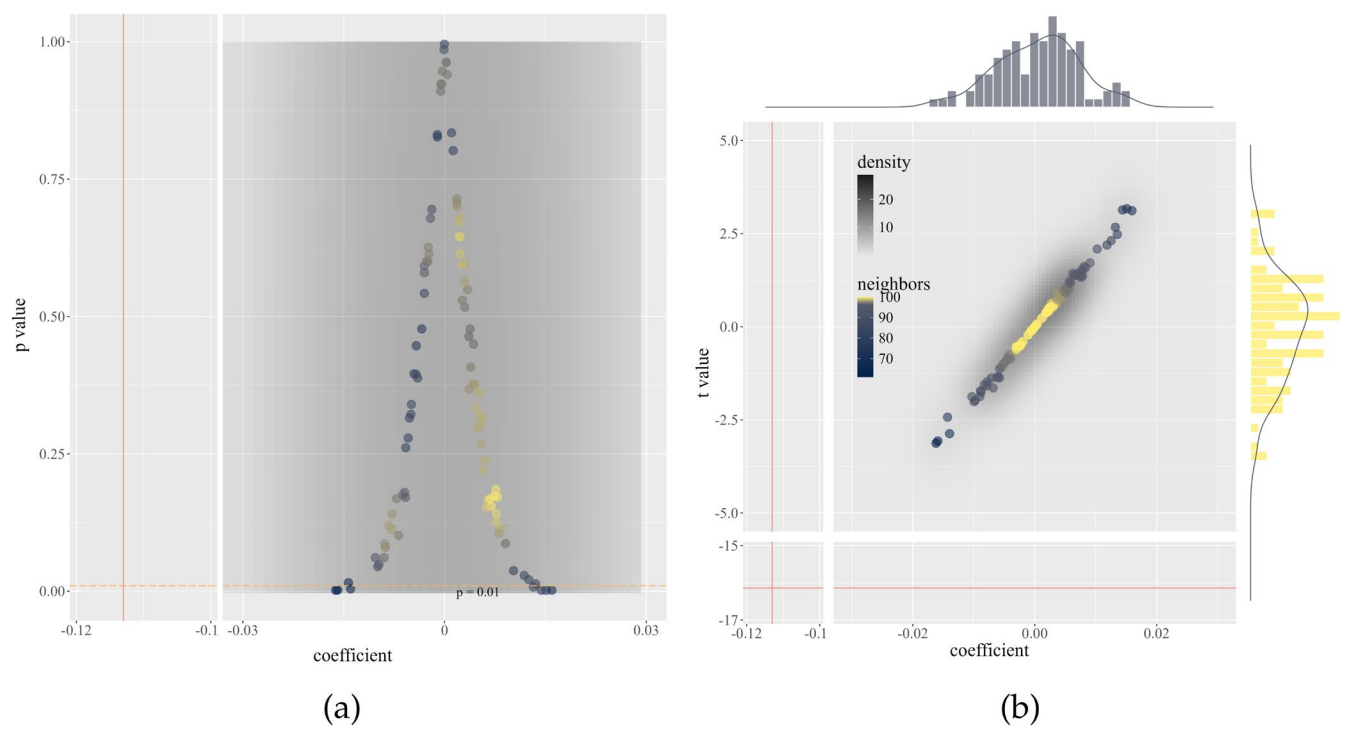


FIGURE 8 | Results of placebo tests. (a) and (b) both exhibit 100 times Bootstrap test results. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

uncertainty and the current reversal of globalization trends, external factors, for example, institutional arrangements among states, geopolitical conditions, and de-globalization pressure, are also likely to shape the relationship between knowledge complexity and economic performance. Therefore, it is necessary to examine further how the BRI interacts with external factors to moderate the effect of complex knowledge on economic growth jointly.

Accordingly, this article introduces the interaction between the BRI and three external moderating variables to investigate their joint moderating effects on the relationship between complex knowledge and economic growth.

In terms of political proximity, this article employs the Ideal Point proposed by Bailey et al. (2017), which introduces spatial-based Dynamic Item Response Theory (IRT). The Ideal Point captures the alignment of countries' foreign policy preferences based on United Nations General Assembly voting patterns.¹³

For geopolitical stability, this article adopts the Geopolitical Risk Index (GPRI) developed by Caldara and Iacoviello (2022), which quantifies the exposure of countries to geopolitical threats based on keyword frequency analyses of major news media.¹⁴

Last, to measure the degree of de-globalization, this article uses weighted average tariffs as a proxy for trade protectionism, as tariff barriers are known to restrict cross-border factor flows and thereby indirectly hinder innovation and technology diffusion. Results under the 0.5 quantile are shown in Table 12; full results of other quantiles are detailed in Tables S39–S41.

The results in the table show that the coefficients of the interaction terms, between *GPYC_rank* and both *ideal* and *geopolit*, are significantly negative. It suggests that greater political alignment (i.e., a higher value of ideal point) and lower geopolitical risk (i.e., a more stable geopolitical environment) strengthen the positive effect of complex knowledge on economic growth. Furthermore, the two triple interaction terms involving the BRI are also significantly negative at sufficiently high levels. It indicates that the BRI plays a “leverage effect” role, further amplifying the growth-enhancing effects of complex knowledge under conditions of political alignment and geopolitical stability.

On the other hand, the coefficient of the interaction term between *wtar* and *GPYC_rank* is significantly positive, implying that a more isolated external environment or more aggressive protectionist policies (i.e., higher tariffs) tend to weaken the

TABLE 12 | Results of moderating effect interacting with ideal point.

Variable	Economic growth					
	Ideal point		Geopolitical risk		Tariff	
	(1)	(2)	(3)	(4)	(5)	(6)
GPYC_rank	−0.105*** (0.003)	−0.117*** (0.004)	−0.046*** (0.007)	−0.097*** (0.011)	−0.177*** (0.003)	−0.173*** (0.003)
GPYC_rank×ideal	−0.002 (0.002)	0.007** (0.003)				
GPYC_rank×ideal×BRI		−0.044*** (0.006)				
GPYC_rank×geopolit			−0.103*** (0.009)	−0.063*** (0.013)		
GPYC_rank×geopolit×BRI				−0.089*** (0.019)		
GPYC_rank×wtar					1.025*** (0.149)	2.400*** (0.193)
GPYC_rank×wtar×BRI						−2.155*** (0.266)
Control variable	✓	✓	✓	✓	✓	✓
Observation	1506	1506	1770	1770	1738	1738
Year FE	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓
R ₁	0.562	0.576	0.599	0.696	0.687	0.701

Note: **, *** denote $p < 0.05$, < 0.01 .

positive impact of complex knowledge on economic growth. Similarly, the triple interaction term involving BRI is significantly negative, suggesting that BRI helps mitigate the adverse effects of de-globalization and enables participating countries to more effectively harness the economic benefits of complex knowledge production.

7 | Concluding Remarks

This article measures and analyzes the knowledge complexity based on the FC algorithm and matrix-estimation exercise within a bipartite network framework; and then, this article conducts a series of empirical studies on the relationship between knowledge complexity and economic growth based on IV-FEQR. Results show that:

First, the calculation and analysis of knowledge complexity reveal that, for a country (or region), knowledge complexity manifests as irrelevance to the total volume of patent outputs; knowledge complexity just depends on the degree of similarity of patent contribution structure among countries, as well as the level of specialization in various technology domains.

Second, the empirical research results indicate: (1) Higher knowledge complexity significantly promotes the level of national economic growth and shows a noticeable temporal continuity. (2) The proposal of the “Belt and Road” can significantly positively moderate the promoting effect of knowledge complexity on economic growth for participating countries. (3) The promoting effect of knowledge complexity on economic growth can be achieved through macroeconomic approaches and innovative practices. The positive impact of complex knowledge production on the economy is more pronounced in technology domains with higher knowledge complexity and in nonlabor-intensive economic sectors with higher value added. (4) The promotion of knowledge complexity on economic growth has significant regional heterogeneity, and the different “expirations” of patents will affect the promotion effect of knowledge complexity significantly, meaning that the outputting of complex knowledge within a short term will have a more pronounced impact on economic growth. Moreover, closer political alliances and more stable geopolitical environments enhance the growth-promoting effects of complex knowledge. Building on these conditions, the Belt and Road Initiative further amplifies such positive effects. Although the trend of de-globalization and rising trade protectionism weakens the economic benefits of complex knowledge production, the BRI nonetheless contributes to mitigating the negative impact of “decoupling” risks under de-globalization, thereby facilitating the effective realization of knowledge-driven growth.

Based on the results, the following insights can be gained:

First, for a country aiming to enhance its economic development by increasing knowledge complexity, achieving the “division of labor” across various technology domains is undoubtedly the best approach. Each country should choose which knowledge fields to engage in or specialize in based on its actual development status and factor endowments to effectively enhance its knowledge complexity.

Second, the promoting effect of knowledge complexity on economic growth is significant and effective. Therefore, countries should actively strive to support complex knowledge production, accumulation, investment, and re-production through policies that support R&D, create a favorable atmosphere for innovation, and build incentive mechanisms. It will foster continuous innovation and iteration across technology domains, thereby promoting rapid economic development.

Third, establishing multilateral cooperation mechanisms, such as the “Belt and Road” Initiative, is highly beneficial. To achieve faster and higher-quality development, countries should jointly discuss and build multilateral cooperation initiatives to foster collaboration in technology domains, which will allow them to benefit more from the cross-border flow of knowledge elements and the spillover effects of complex knowledge, thereby achieving mutual benefit and win-win outcomes.

Furthermore, in pursuing higher economic growth, countries should focus on continuous and abundant technological innovation and output in advanced, high-tech fields that signify more complex knowledge, rather than merely relying on mature technological industries with completed technological accumulation. In fact, the continuous and leading technological innovation in less explored fields is the true source of economic growth in the short term.

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The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Endnotes

¹ With the advancement of globalization, it is well recognized that knowledge is indispensable in the production process (Dunning 2000). The competitiveness of firms or economic entities depends on the production of high-value, non-ubiquitous, complex, and tacit knowledge (Asheim and Gertler 2006). Currently, with economic globalization setbacks, the disparity between regions in the generation and dissemination of knowledge plays a more noticeable role in economic growth and development imbalance (Corrado and Hulten 2010).

² Although existing literature has extensively discussed the economic impacts of complex knowledge across various fields, such as economic growth (Arthur 1999; Audretsch and Belitski 2021; Balland et al. 2019; Helbing and Kirman 2013; Maradana et al. 2017), employment (Eriksson and Forslund 2014; Ganco 2013), industrial upgrading (Bai et al. 2020; Feng and Liu 2018), innovation (Audretsch and Belitski 2021; Fischer and Frhlich 2010; Ruoslahti 2020), and international competitiveness (Feng and Liu 2018; Tacchella et al. 2013), there still remains considerable room for expansion in the measurement of knowledge complexity, especially for the heterogeneity of knowledge and the correlation between research objects.

- ³ An exception is the case of CO₂ emissions, where the magnitude of the coefficient becomes significantly larger (in absolute terms) at higher quantiles. This may be explained by the tendency of less developed countries (with higher GPYC_rank values) to pursue industrial growth strategies that rely heavily on carbon-intensive activities, thereby sacrificing environmental sustainability.
- ⁴ High-level diplomatic relationships or partnerships are defined as co-operative or strategic partnerships titled with “comprehensive,” “all-weather,” “all-round,” or “global.” See Table S13.
- ⁵ The notion of a “Community of Shared Destiny” builds upon high-level strategic partnerships and entails broader and deeper cooperation. It also reflects an endorsement of China’s normative vision of constructing a “community with a shared future for mankind.” A complete list of such countries is provided in Table S12.
- ⁶ Participation in the BRI is generally formalized through the signing of bilateral cooperation documents, which typically proceed in stages—from letters of intent, to MoUs or framework agreements, and finally to binding contracts or treaties. The timeline of BRI participation of countries is listed in Table S10.
- ⁷ The AIIB was created to promote regional connectivity and economic integration in Asia and to support infrastructure development among BRI countries. Its prospective founding members include 37 regional and 20 nonregional countries. A full list of founding members is provided in Table S11.
- ⁸ For example, Electrical machinery, apparatus, energy (WIPO.1), Audio-visual technology (WIPO.2), Telecommunications (WIPO.3), Digital communication (WIPO.4), Computer technology (WIPO.6), Semiconductors (WIPO.8), Optics (WIPO.9), Micro-structural and nano-technology (WIPO.22), etc.
- ⁹ For example, Organic fine chemistry (WIPO.14), Biotechnology (WIPO.15), Pharmaceuticals (WIPO.16), Food chemistry (WIPO.18), Environmental technology (WIPO.24), Mechanical elements (WIPO.31), Civil engineering (WIPO.35), etc.
- ¹⁰ For example, Fishing and aquaculture (03), Mining support service activities (09), Chemical and chemical products (20), etc.
- ¹¹ Within PSM, the article will choose caliper-nearest as the matching method and generalized linear model as distance measurement. Then, IVFEQRs are conducted based on the matched samples. Before conducting the formal regressions, an imbalance test is performed on the matched sample data. Tables S29–S37 shows the results of the sample based on caliper-nearest matching, which indicates that the matched samples passed the test, with no significant differences between all covariates in the two groups.
- ¹² Figure 7 shows the parallel test result with GDPpc_growth as the explained variable; results of the others are shown in Figure S2.
- ¹³ Here, China is designated as the reference country, and the spatial positions of all other countries or regions are recalculated relative to China’s proposals. This yields updated estimates of the Ideal Point. Setting China as the reference point implies that a higher estimated Ideal Point value indicates a greater alignment of a country’s political position with that of China.
- ¹⁴ Here, this article applies the reverse Max-Min standardization method to GPRI to measure geopolitical stability.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** rode70156-sup-0001-supinfo.pdf. **Appendix S1:** Knowledge complexity measurement. **Appendix S2:** GPYT and GPYC results. **Appendix S3:** Baseline regression results and robustness checks. **Appendix S4:** Mechanism tests and heterogeneous analyses. **Appendix S5:** Moderating role of the "Belt and Road" Initiative. **Table S1:** Full GPYT results of technology domains in 2021. **Table S2:** Full GPYC results of 100 countries in 2021. **Table S3:** Abbreviations of variables and corresponding names and definitions. **Table S4:** Abbreviations of variables and corresponding names and definitions. **Table S5:** Descriptive statistics of each variable for robustness checks. **Table S6:** Descriptive statistics of each variable for Global sample and moderating effect model. **Table S7:** Result of first stage of baseline regressions. **Table S8:** Results of baseline regressions with GDPpc_growth as explained variable. **Table S9:** Results of baseline regressions with GNipc_growth as explained variable. **Table S10:** Timelines of countries joining the Belt and Road Initiative. **Table S11:** All prospective founding members of AIIB. **Table S12:** Countries of Bilateral Community of Common Destiny. **Table S13:** Countries and their bilateral diplomatic relationship with China. **Table S14:** Robustness checks results based on household consumption. **Table S15:** Robustness checks results based on CO₂ emission calculated via consumption. **Table S16:** Robustness checks results based on Comprehensive Wealth per capita. **Table S17:** Robustness checks results based on Human Capital per capita. **Table S18:** Robustness checks results based on nighttime light. **Table S19:** Robustness checks

results based on electricity production. **Table S20:** Robustness checks results based on rail lines. **Table S21:** Results of samples grouped by charges for intellectual property. **Table S22:** Results of samples grouped by industrial optimization. **Table S23:** Results of samples grouped by R&D practice. **Table S24:** Results of samples grouped by R&D expenditures. **Table S25:** Results of samples grouped by geographical or ethno-cultural region. **Table S26:** Results of samples grouped by "expiration" of patents. **Table S27:** Industries in OECD ICIO table. **Table S28:** 97.5% confidence intervals of two samples. **Table S29:** Results of covariates imbalance tests (Dependent variable: GDPpc_growth). **Table S30:** Results of covariates imbalance tests (Dependent variable: GNipc_growth). **Table S31:** Results of covariates imbalance tests (Dependent variable: household consumption). **Table S32:** Results of covariates imbalance tests (Dependent variable: CO₂ emission based on consumption). **Table S33:** Results of covariates imbalance tests (Dependent variable: Comprehensive Wealth per capita). **Table S34:** Results of covariates imbalance tests (Dependent variable: Human Capital per capita). **Table S35:** Results of covariates imbalance tests (Dependent variable: nighttime light). **Table S36:** Results of covariates imbalance tests (Dependent variable: electricity production). **Table S37:** Results of covariates imbalance tests (Dependent variable: rail lines). **Table S38:** Moderating effect results of Belt and Road with different aspects of economic development. **Table S39:** Results of moderating effect interacting with Ideal Point. **Table S40:** Results of moderating effect interacting with geopolitical risk. **Table S41:** Results of moderating effect interacting with Weighted Average Tariff. **Figure S1:** Full results of baseline regressions. **Figure S2:** Results of Parallel tests.